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Research article

Can ChatGPT replace humans in crisis communication? The effects of AI-mediated crisis communication on stakeholder satisfaction and responsibility attribution

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ABSTRACT

Imagine a world where chatbots are the first responders to crises, efficiently addressing concerns and providing crucial information. ChatGPT has demonstrated the capability of GenAI (Generative Artificial Intelligence)-powered chatbots when deployed to answer crisis-related questions in a timely and cost-efficient manner, thus replacing humans in crisis communication. However, public reactions to such messages remain unknown. To address this problem, this study recruited participants ($N_1 = 399$, $N_2 = 189$, and $N_3 = 121$) and conducted two online vignette experiments and a qualitative survey. The results suggest that, when organizations fail to handle crisis-related requests, stakeholders exhibit higher satisfaction and lower responsibility attribution to chatbots providing instructing (vs. adjusting) information, as they are perceived to be more competent. However, when organizations satisfy requests, chatbots that provide adjusting (vs. instructing information) lead to higher satisfaction and lower responsibility attribution due to higher perceived competence. The second experiment involving a public emergency crisis scenario reveals that, regardless of the information provided (instructing or adjusting), stakeholders exhibit greater satisfaction and positive attitudes toward high-competence (vs. low-competence) chatbots. The qualitative study further confirms the experimental findings and offers insights to improve crisis chatbots. These findings contribute to the literature by extending situational crisis communication theory to nonhuman touchpoints and providing a deeper understanding of using chatbots in crisis communication through the lens of machine heuristics. The study also offers practical guidance for organizations to strategically integrate chatbots and human agents in crisis management based on context.

1. Introduction

During crises, every second counts, and the ability to communicate clearly and efficiently is paramount. Traditional crisis communication typically involves human spokespersons relaying information to stakeholders. However, rapid advancements in chatbot technology have demonstrated a remarkable capability in processing language and providing accurate, relevant response (e.g., Dwivedi et al., 2023; Hofeditz et al., 2019; Stieglitz et al., 2022). In particular, the introduction of ChatGPT (Chat Generative Pretrained Transformer) has demonstrated the immense capability of GenAI (Generative Artificial Intelligence)-powered chatbots (Dwivedi et al., 2023). Built on OpenAI's GPT large language models (LLMs), ChatGPT can accurately answer users' questions and explain relevant information in depth (Sundar, 2023)). Given that chatbots can be programmed to answer a wide range of questions and provide detailed information on specific topics, which can be useful in situations in which people need to access information quickly and easily, questions have emerged as to whether ChatGPT and similar AI technologies offer suitable replacements for human communication during crises (Cheng & Jiang, 2020; Chin et al., 2023; Hofeditz et al., 2019; Maniou & Veglis, 2020). Chatbots can quickly process large amounts of data and provide responses in real time, which is particularly helpful during crises, when time is of the essence. Furthermore, chatbots can handle multiple interactions simultaneously, thus benefiting crisis situations wherein many people need information at the same time.

While researchers have acknowledged the potential benefits of AI in the field of public relations (Galloway & Swiatek, 2018), limited scholarly attention has been given to the use of chatbot technology in

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crisis communication in recent years. To date, humans have been the primary creators of crisis messaging, with theoretical frameworks, such as the situational crisis communication theory (SCCT, Coombs, 1995; Coombs and Holladay, 2002) and the contingency theory of accommodation (Pang et al., 2020), focusing on understanding accelerators and barriers to human efforts in crisis management. Despite updates to crisis communication knowledge prompted by the widespread use of social media among public relations practitioners (e.g., the revised SCCT, Coombs, 2017; social-mediated crisis communication (SMCC), Jin & Liu, 2010), these frameworks still view traditional and social media as the only channels for conveying human decisions. With the advancement of AI, chatbots have emerged as machine-based simulations or replications of human intelligence via machine learning (ML) algorithms that learn over time by recognizing patterns in data (Stewart et al., 2020). Consequently, chatbots no longer serve as mere mediums, but have become spokespersons for organizations in their interactions with stakeholders. Nevertheless, it remains uncertain whether existing crisis communication theoretical frameworks are still applicable to chatbots as sources of information.

The current state of research on chatbots in crisis communication highlights some promising yet underexplored areas. For example, while studies indicate that chatbots can effectively deliver timely and relevant information during crises (Hofeditz et al., 2019; Stieglitz et al., 2022), there is still much to learn about how specific types of crisis responses (i.e., instructing vs. adjusting) are perceived by the public.

Another important factor that may moderate the effectiveness of chatbots in crisis communication is the crisis handling status - whether the chatbot is able to successfully address and resolve the stakeholder's crisis-related request or not. Prior research in the marketing domain has shown that people prefer AI agents when outcomes are worse than expected, due to the perceived lack of selfish intentions of AI (Garvey et al., 2023). In addition, consumers are more likely to be more satisfied with a chatbot agent than a human agent if their service request is not satisfied (Yu et al., 2022) Given the unpredictable nature of crises, an agent's response may either satisfy or dissatisfy stakeholders' requests (Coombs, 2007). However, whether this preference pattern holds in crisis communication remains unclear. Addressing this gap is critical to optimize chatbot deployment in crisis management.

In addition, the competence of the spokesperson—whether AI or human—plays a crucial role in stakeholders' confidence in an organization's ability to manage crises, ultimately affecting its reputation (Coombs, 2015; James & Wooten, 2005). In turn, this raises questions on how the public perceives the comparative competence of AI and human spokespersons in a crisis context. Understanding public perceptions of AI-mediated crisis communication is essential in evaluating the potential of chatbot technology in emergency response scenarios. Furthermore, insights into this area can help determine whether stakeholders are satisfied with chatbots providing information during crises and whether such technologies could be widely adopted. To address these gaps, this paper proposes the following research questions:

RQ$_1$: What impact do chatbots (vs. humans) have on stakeholder satisfaction and responsibility attribution during crises?

RQ$_2$: Does the type of information (instructing vs. adjusting) influence the effectiveness of AI-mediated crisis communication?

RQ$_3$: Does the effectiveness of AI-mediated crisis communication differ when chatbots fail to satisfy stakeholders' requests?

RQ$_4$: What role does perceived competence play in AI-mediated crisis communication?

RQ$_5$: How does the public perceive the value or threat of AI-mediated crisis communication?

The answers to these questions can deepen our understanding of how AI can be optimally used in crisis communication and how it is perceived by the public in terms of satisfaction, responsibility, and competence. To answer these questions, this paper conducts two experimental studies and a qualitative study to comprehend public reactions to crisis communication when it is mediated by AI, particularly chatbots, as opposed to human agents. Moreover, this work explores whether the nature of the information provided (i.e., instructing or adjusting) and the ability of such information to satisfy stakeholders have an impact on chatbots' perceived competence, subsequently influencing stakeholder satisfaction and responsibility attribution.

The findings of this study advance our understanding of SCCT by extending its application to AI-mediated interactions, marking a significant contribution to the crisis communication literature by examining nonhuman entities as information disseminators. This research also enriches the machine heuristic literature by demonstrating how chatbots—perceived through the lens of machine heuristics—can effectively manage crisis information when delivering instructive information. By demonstrating how perceived competence mediates stakeholder satisfaction and responsibility attribution in the context of chatbot communications, this study provides a better understanding of AI's capabilities and limitations in crisis scenarios, thereby offering valuable insights for integrating chatbots into strategic crisis communication frameworks.

The subsequent sections of this paper are structured to provide a comprehensive exploration of the research topic. Following this introduction, the Literature Review section focuses on existing studies on AI-mediated communication and crisis management, setting the stage for a deeper understanding of the research context. The subsequent sections outline the research methodology, present an analysis of the findings, and discuss the implications of the results in the broader context of AI in crisis communication. Finally, the paper concludes by summarizing key insights, acknowledging limitations, and suggesting future research directions.

2. Literature review

2.1. Situational crisis communication theory

Compared with previous case-based crisis communication theories, such as Benoit's (1997) image restoration theory , SCCT offers a more empirical approach to linking crisis types with appropriate crisis response strategies (Coombs, 2007). SCCT is grounded in Weiner's (1972) attribution theory, which posits that individuals tend to attribute responsibility for incidents to certain agents. Crisis responsibility, defined as "the degree to which stakeholders attribute responsibility to an organization" (Coombs, 2004, p. 268), plays a central role in SCCT (Coombs, 2007). In particular, this theory identifies three crisis types based on the level of responsibility attributed to the organization: the victim crisis, the accidental crisis, and the preventable crisis clusters (Coombs, 2007; Coombs & Holladay, 2002). In a victim crisis, such as natural disasters, an organization is also considered a victim and is thus attributed low responsibility. In an accidental crisis, such as technical errors, an organization is attributed limited responsibility because the harm caused is unintentional. In a preventable crisis, such as organizational misdeeds, an organization bears strong responsibility if it puts stakeholders at risk through avoidable actions.

Crisis response messages are categorized into three types: instructing information, adjusting information, and reputation repair strategies (Coombs, 2015; Sturges, 1994). On the one hand, the instructing and adjusting types of information prioritize the physical and psychological needs of victims or potential victims. On the other hand, reputation repair strategies focus on addressing concerns about the organization's reputation and rebuilding it postcrisis. SCCT further divides reputation repair strategies into three categories based on the level of responsibility accepted: denial (e.g., denial and scapegoating), diminish (e.g., excusing and justification), and rebuild (e.g., apology and compensation) (Coombs, 2007). SCCT emphasizes the importance of aligning the level of crisis responsibility with the crisis response strategy's perceived acceptance of responsibility. In this case, effective crisis communication should match the situational aspects of crisis responsibility to mitigate the reputational threats posed by a crisis to an organization.

Instructing and adjusting information can be considered base responses (Coombs, 2021). This notion suggests that crisis responses should initially address these types of information before shifting the focus to reputation repair strategies, particularly when there are victims or potential victims (Coombs, 2022; Sturges, 1994). In some instances, only the base response may be adequate for reputation repair, especially when perceived crisis responsibility is low (Coombs, 2017). In actual practice, many preventable crises can be managed without any reputation-repair strategies (Kim and Sung, 2014; Park, 2017). Furthermore, Page's (2022) findings suggest that solely using base crisis responses may have a more significant impact on reputation repair than the reputation repair strategies recommended by SCCT. However, compared with a multitude of studies investigating the effectiveness of reputation repair strategies in crisis communication (e.g., Coombs, 2015, 2022; Xiao et al., 2018), limited research has been conducted to fully explain the process of employing a base crisis response to repair an organization's reputation. Moreover, the effectiveness of chatbots in delivering base crisis responses and how this attribute compares to human-delivered crisis responses remains a relatively unexplored area in the field of crisis communication.

2.2. The use of chatbots vs. human agents in crisis communication

A growing number of studies (e.g., Holthöwer & van Doorn, 2022; Yu and Zhao, 2024) have focused on users' different perceptions regarding human vs. chatbot agents (see Table 1 for an overview). Although these studies use different terms, such as "AI," "chatbot," "robot," "bot," "algorithm," "machine," "computer," or "automation," the nature of the technology is similar. These terms represent computer programs that use AI algorithms to respond automatically to user input. In this study, we use the term "chatbot" to refer to such an AI-powered automatic response system.

Previous studies (e.g., Tay et al., 2014; Yu and Zhao, 2024)) that examine human–chatbot interactions are mainly built on the "computers as social actor paradigm" (CASA). This paradigm contends that people unconsciously employ the same social heuristics they use in their interactions with other people when dealing with computers because the latter remind them of the same social characteristics (Nass and Moon, 2000). In other words, people tend to respond to a chatbot in the same way that they interact with humans, and such an effect is often an outcome of a process of mindlessness (Araujo, 2018; Kim & Sundar, 2012; Langer, 1992; Nass & Moon, 2000). Several studies have documented evidence of CASA in human–computer interactions (e.g., Tay et al., 2014; Yu & Zhao, 2022). For example, people prefer a female (vs. male) healthcare robot and a male (vs. female) security robot (Tay et al., 2014), despite the "gender" of the robot being meaningless to the task performance. Anthropomorphism is often used to activate the social presence of chatbots. For example, people react to healthcare chatbots more positively if the chatbots express emotions (Yu & Zhao, 2022).

However, a growing number of studies also suggest that people interact with chatbots and humans differently (e.g., Loughnan & Haslam, 2007; Yu et al., 2022; Yu and Zhao, 2024) because chatbots possess a number of unique features compared to humans. Humans have autonomy, motivations, emotions, biases, and sensations, whereas bots do not (Gray & Wegner, 2012; Loughnan & Haslam, 2007). In comparison, bots are programmed to follow a certain set of rules to complete tasks (Monroe et al., 2014; Yogeeswaran et al., 2016). These differences can result in a phenomenon called "machine heuristics," which refers to the mental shortcuts that people use when interacting with and evaluating chatbots. As a result, consumers attribute less agency and responsibility to algorithms compared to humans (Srinivasan & Sarial-Abi, 2021). Due to less agency and limited flexibility, we have lower expectations of chatbots (Crolic et al., 2022; Yu et al., 2022; Zhou et al., 2023).

Although the lack of flexibility in chatbots results in the so-called algorithm aversion (Fryer et al., 2017; Luo et al., 2019; Mende et al., 2019; Önkal, Goodwin, Thomson, Gönül, & Pollock, 2009; Promberger & Baron, 2006), it also makes service failure more acceptable for chatbot agents compared to human ones (Mozafari et al., 2022; Yalcin et al., 2022; Yu et al., 2022) because people tend to have a lower expectation of flexibility for bots. As a result, chatbots are perceived to be more objective. An example of the "chatbots being objective effect" is that we trust chatbots more than humans with our personal information (Sundar & Kim, 2019).

However, if customers are in an angry emotional state when they initiate a chatbot-led service interaction, the attribution of anthropomorphism qualities to the chatbot has an adverse impact on their

Table 1
Relevant literature on people's perceptions of humans vs. chatbots.

Topic	Theoretical Lens	Method	Main Findings	Source
Chatbots and Emotional Expression	CASA theory	Experiment	Chatbots expressing emotions can shorten the psychological distance between the user and the chatbot, enhancing user experience and engagement.	Yu & Zhao (2022)
Service Robots and Embarrassing Encounters	Robots' lack of judgment (Machine heuristic)	Experiment	Consumers prefer interacting with robots over humans in embarrassing service encounters because they feel less judged. This preference facilitates acceptance of the robots' service and reduces consumers' reluctance to accept service offerings in embarrassing situations.	Holthöwer & van Doorn (2022)
Marketing and Consumer Response	AI's lack of intention (Machine heuristic)	Experiment	Consumers perceive AI agents as having weaker intentions than humans, thus influencing their responses to negative and positive offers. This perception leads to moderated responses to AI-delivered offers, which in turn, affect purchase likelihood and satisfaction.	Garvey et al. (2023)
Consumer Reactions to Rejection	Inflexibility of chatbots (Machine heuristic)	Experiment	Consumers exhibit less dissatisfaction when service rejections are delivered by robots than by humans, which is likely due to lower expectations of flexibility from chatbots.	Yu et al. (2022)
Algorithmic vs. Human Decision-Making	Attribution processes	Experiment	Consumers react less positively to favorable decisions made by algorithms than those made by humans, while reactions to unfavorable decisions are similar regardless of the decision-maker. This difference is attributed to the ease of internalizing favorable outcomes from humans and externalizing unfavorable outcomes regardless of the decision-maker.	Yalcin et al. (2022)
Emoji Usage	Artificial emotions	Experiment	Emojis sent by chatbots are perceived as less persuasive and genuine than those sent by humans, indicating a qualitative difference in how artificial and human-expressed emotions are received by users.	Yu and Zhao, 2024
Brand Harm Crises and Algorithm Failures	Theory of mind perception	Experiment	Following a brand harm crisis, consumers place less blame on algorithms than humans, attributing lower agency and responsibility to the former. Anthropomorphism and task nature moderate this effect. This study offers insights into managing algorithm-caused crises and mitigating negative consumer reactions.	Srinivasan & Sarial-Abi (2021)

satisfaction (Crolic et al., 2022). Liu-Thompkins et al. (2022) further explained that the reason why anthropomorphism has a negative impact in angry situations could be attributed to the inconsistency between a human-like appearance and a lack of empathy. Therefore, Garvey et al. (2023) suggested that when there is bad news, it is better to send an AI because it is perceived to lack intention. Furthermore, people tend to prefer a chatbot to a human for embarrassing service encounters because bots do not judge (Holthöwer & van Doorn, 2022; Sun et al., 2023). However, due to their lack of emotional capability, bots using emojis during a conversation are less effective than when humans do the same (Yu & Zhao, 2024).

Although the different perceptions of chatbot vs. human service providers have been studied extensively in recent years, studies on how people react to chatbots in the context of crisis communication are scarce. It remains unknown whether chatbots can effectively replace humans in crisis communication. On the one hand, people may trust crisis-related information delivered by chatbots due to machine heuristics (Sundar & Kim, 2019). On the other hand, crisis responses not only include factual information and guidelines but also emotional support. To understand this, we first need to understand the different crisis communication strategies.

3. Hypotheses development

3.1. The moderating role of instructing and adjusting information in crisis communication

Through their verbal cues, spokespersons play a crucial role in shaping public perceptions of an organization during a crisis (Claeys & Cauberghe, 2014; Li et al., 2023). When it comes to verbal cues, chatbots may excel in communicating instructing crisis messages but may struggle with adjusting crisis messages compared to humans.

Instructing information refers to messages that inform stakeholders about how to protect themselves physically during a crisis. These messages can satisfy stakeholders' safety needs to understand the effects of the crisis and what actions they should take to protect themselves physically (Coombs, 2021). Instructing information can also be critical in emergency situations where time is of the essence, as it can help stakeholders take immediate actions to protect themselves and restore order from the chaos invoked by a crisis (Sellnow et al., 2012). This information can be especially important in cases of natural disasters, hazardous incidents, or other types of emergencies where timing is critical. In addition, providing instructing information can be an effective way for organizations to demonstrate their concern for the well-being of their stakeholders and their efforts to mitigate a crisis. Examples of instructing information include warnings and directions to help stakeholders stay safe and avoid physical harm, such as evacuation maps, shelter-in-place instructions, and information on how to return defective products (Sturges, 1994).

Adjusting information refers to messages that help stakeholders cope psychologically with crises. Crises are often information-poor situations that can be stressful for stakeholders due to uncertainty and potential harm. To alleviate this stress, stakeholders need to receive as much information as possible to be reassured (Coombs, 2021). Therefore, information about what happened and what corrective actions are being taken to prevent a similar crisis in the future is crucial for coping with psychological stress during a crisis (Sellnow et al., 1998). However, corrective actions can take a long time to develop until the cause of a crisis is determined.

Recent research has emphasized that adjusting information should include an organization's emotional response (i.e., affective empathy) to the victims' situation as the main component of an ethical base response. "Empathy" refers to one's sensitivity and responsibility to others (Hartup, 1993), and showing affective empathy can help alleviate stakeholders' suffering during crises by considering their perspectives of the situations at hand (Schoofs et al., 2022). For example, this may include showing understanding of stakeholders' experiences and expressing sadness (Schoofs and Claeys, 2021), experiencing anger and hope (Xiao et al., 2018), and highlighting a commitment to addressing their needs and concerns (Schoofs et al., 2022). Therefore, the content of adjusting information should consist of three types of messages: explaining stakeholders' concerns about what happened, expressing empathy for victims, and describing the organization's corrective actions (Claeys et al., 2022; Coombs, 2022).

Base responses are typically presented as a combination of instructing and adjusting information, as both can reduce the uncertainties created by crisis situations (Holladay, 2009; Sturges, 1994). However, these types of information are increasingly being discussed as separate categories to be evaluated and compared. For instance, in a field experiment on a health crisis, providing comprehensive instructing and adjusting information was found to be more effective than solely offering limited (mainly adjusting) information during the acute phase of a crisis (Claeys et al., 2022). According to a revised model of reputation repair (REMREP), providing instructing information (vs. adjusting information) increases the perception of the virtuousness of the organization in crisis, which has a greater impact on organizational reputation (Page, 2022). Moreover, Kim et al. (2023) revealed that providing employees with instructing information about COVID-19 led to increased trust in the organization's handling of the pandemic. In turn, this resulted in employees showing greater support for organizational decisions, feeling a stronger sense of connection to the organization, and having higher satisfaction in their relationship with the organization. In comparison, providing adjusting information had a negative impact on employees' trust in the organization's commitment. In the current study, we also investigated the use of chatbots versus humans as a means of disseminating a base crisis response for reputation repair. This is done by examining the separate roles of instructing and adjusting information.

Chatbots possess distinct features, such as a lack of emotions, motivations, biases, or sensations, that set them apart from human beings (Gray & Wegner, 2012; Loughnan & Haslam, 2007). Still, people perceive AI to be as equally fair and trustworthy as humans in terms of assigning work-scheduling tasks (Lee, 2018) and prefer AI over humans for tasks with objectively correct solutions (Logg, Minson, & Moore, 2019) . Moreover, users exhibit greater acceptance of chatbot service failure (Mozafari et al., 2022; Yu et al., 2022) due to lower expectations of chatbots' adaptability (Yu et al., 2022). Chatbots are also perceived to be more objective than humans (Sundar & Kim, 2019) because they lack biases or prejudices (Holthöwer & van Doorn, 2022). Such attributes may make them suitable for disseminating crisis-related instructive information, which often requires reliability, consistency, and standardization (Stieglitz et al., 2022).

However, during crises, organizations must also provide adjusting information that involves emotional support and empathy to reassure stakeholders (Coombs, 2021). These tasks are more complex and nuanced, potentially requiring human intervention. Human spokespersons during crises can express emotions through eye contact, facial expressions, and expressive body movements (Claeys & Cauberghe, 2014). In comparison, due to AI's lack of emotion capability, emotions expressed by chatbots are viewed as less authentic and persuasive (Yu and Zhao, 2024). Therefore, chatbots may not outperform humans in terms of providing adjusting information.

3.2. The moderating effect of crisis-handling status in chatbot communication

Drawing on the machine heuristic literature, we further expect the aforementioned effect to depend on the crisis-handling status. Here, "crisis-handling status" refers to the outcome of how an organization handles a crisis-related request. For example, in a product-harm crisis, an organization may refuse consumers' requests to return the product and receive monetary compensation. We posit that chatbots may have an advantage over humans when providing clear and consistent instructing information if consumers' requests are rejected. This is because people tend to have a lower expectation of chatbots' flexibility in handling a crisis (Yu et al., 2022).

Chatbots are programmed to follow a set script, which means that they can provide consistent and accurate information without the risk of deviating from the intended message (Monroe et al., 2014; Yogeeswaran et al., 2016). Prior studies have shown that people perceive AI to be as fair and trustworthy as humans for certain tasks, such as work assigning and scheduling (Lee, 2018), as well as for numerical tasks with objectively correct answers (Logg et al., 2019). Therefore, in receiving instructing information under a failed crisis-handling status, people tend to trust chatbots more than humans. However, when it comes to providing adjusting information, such as emotional support and empathy, chatbots may not be better than human beings (Castelo et al., 2019). Although prior research has found that the use of emojis can increase consumer forgiveness in online service recoveries (Wang et al., 2023), the negative effect of emotional disclosure on service failure has also been observed. For instance, when chatbots fail to handle consumers' service requests, consumers perceive an emotional apology from chatbots as less sincere than that from humans because the former do not have authentic emotional abilities (Yu et al., 2022). It has generally been acknowledged that chatbots do not have real emotions. When a crisis-related request is rejected, showing emotional support may make this feature more salient because people believe that chatbots may not be able to understand the emotional state with which they are communicating and adjust their response accordingly.

H1. : The effect on competence of using chatbots to replace humans in crisis communication depends on the crisis-handling status and crisis-handling strategy.

H1a. . Using chatbots to replace humans for instructing information (vs. adjusting information) leads to higher competence when agents are unable (vs. able) to solve a crisis-related problem.

However, when a crisis-related request is handled successfully, chatbots' emotional inability may become less salient. In contrast, consumers may perceive adjusting information from chatbots as more competent because it exceeds their expectations. Yu and Zhao (2022) revealed that when consumers' service requests are solved, showing emotions shortens the psychological distance of chatbots. The recent literature on empathic chatbots has focused on developing algorithms that can understand and respond to users' emotional states. A growing number of studies have used ML techniques to train chatbots to recognize and respond to different emotions (e.g., Adikari et al., 2022), while others have focused on incorporating natural language processing (e.g., Pandey et al., 2022) and other AI technologies to improve chatbots' ability to understand and respond to users' emotional cues (e.g., Khalil et al., 2019). Practically, Jiang et al., 2022 found empathic human–AI interaction to help people worldwide cope with the overwhelming psychological distress caused by the COVID-19 pandemic. According to Liu-Thompkins et al. (2022), increasing agent empathy can lead to more comparable affective and social customer experiences between AI-enabled and human-based marketing interactions. When chatbots can satisfy users' needs in a crisis and possess the ability to empathize and tailor their responses to specific needs, users may perceive the chatbots as more competent, as they may not have such high expectations of chatbots nowadays. Accordingly, we propose that chatbots will be more effective than humans in providing adjusting information when the crisis-handling status is successful.

H1b. : Using chatbots to replace humans for adjusting information (vs. instructing information) leads to higher competence when agents are able (vs. unable) to solve a crisis-related problem.

3.3. The mediating role of perceived competence in crisis communication

"Perceived competence" refers to the degree to which stakeholders believe an organization is capable of effectively handling a crisis. Kim et al. (2016) found that employees' perceptions of their organization's competence during a crisis had a positive effect on their reactions to the crisis, including their levels of trust, job satisfaction, and organizational commitment. Similarly, public perceptions of a CEO's competence can have a significant impact on an organization's reputation during an ongoing crisis. In such cases, when stakeholders are uncertain and in need of reassurance, increased perceptions of a spokesperson's competence can positively affect the organization's reputation by implicitly conveying that it is in control of the situation (Coombs, 2015; James & Wooten, 2005). Various techniques have been proposed to improve the appearance of a spokesperson's competence, especially during the acute crisis stage. For instance, powerful nonverbal cues, such as vocal cues (e.g., speaking with a lowered voice pitch) and visual cues (e.g., eye contact, expressive body movements, and relaxed facial expressions), can help increase public perception of an organizational spokesperson's competence and thus minimize reputational damage (Claeys & Cauberghe, 2014). However, expressing emotions such as sadness during a crisis can lower perceptions of the CEO's competence, which in turn, can negatively affect the organization's reputation (Schoofs & Claeys, 2021). In the current study, we anticipate that the revelation of an AI-enabled or human-based agent can affect stakeholders' perceptions of the agent's competence. We expect that this, in turn, will influence stakeholders' responsibility attribution and satisfaction with the organization. Thus, we propose the following hypothesis:

H2. : Perceived competence leads to stakeholders' higher satisfaction and lower responsibility attribution.

H3. : The effect of using chatbots to replace humans in crisis communication on stakeholders' satisfaction and responsibility attribution is mediated by perceived competence.

The study's conceptual framework is shown in Fig. 1.

4. Research design

This research adopts a mixed-method approach comprising three interconnected studies, each designed to unpack different facets of AI-mediated crisis communication and its comparison with human agents. The studies were approved by the Academic Ethics Committee of the University. Study 1, an experimental design (N = 399), aimed to quantitatively assess the impact of chatbots vs. human agents on stakeholder satisfaction and responsibility attribution across varied scenarios of crisis response framing and request handling. By manipulating agent type (chatbot vs. human), message framing (instructing vs. adjusting information), and request handling (success vs. failure), this study tested H1 and H2.

Building on the quantitative findings of Study 1, Study 2 (N = 189) further investigates the specific role of perceived competence, specifically exploring how varying levels of chatbot competence influence stakeholder responses during a crisis. This study is designed to establish the causal relationship between perceived competence and stakeholder satisfaction and stakeholders' attitudes toward chatbot agents, further supporting H2.

Study 3 (N = 121) transitions to a qualitative research design that offers a complementary perspective to the preceding quantitative analyses. This specific study aims to capture the detailed subjective experiences and perceptions of individuals interacting with chatbots during crisis scenarios. By employing a user scenario and open-ended questionnaire responses, this study seeks to uncover the strengths, concerns, and suggested improvements for chatbot use in crisis communication.

Together, these three studies form a coherent research design that not only addresses the efficacy of chatbots in crisis situations, but also provides a holistic understanding of the stakeholder experience, thus guiding future advancements in AI-powered crisis communication.

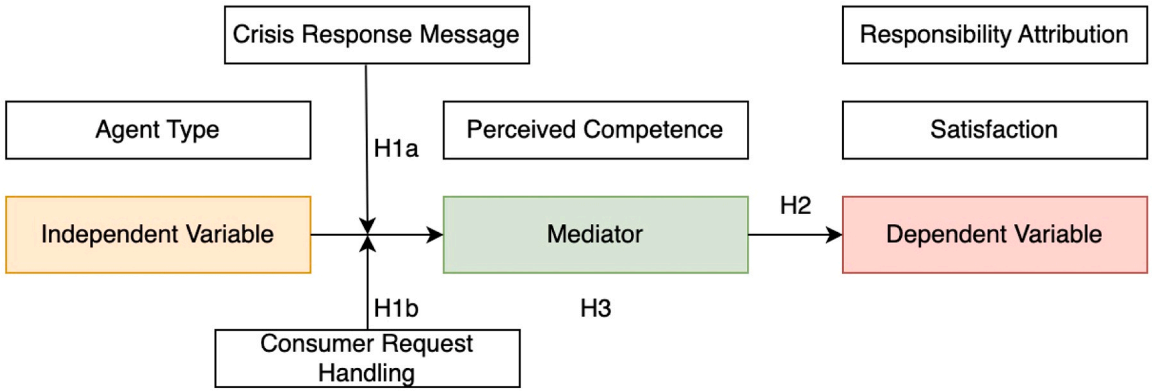

Fig. 1. The hypothesized conceptual model.

5. Study 1: a comparison of chatbots and human agents in crisis communication

5.1. Methods

5.1.1. Study design and stimuli

We conducted an online experiment with a 2 (online agent: chatbot vs. human agent) × 2 (crisis response framing: instructing vs. adjusting information) × 2 (request handling: success vs. failure) between-subjects design in China.

The vignette used in the experiment is related to the crisis communication of a fictitious mobile phone maker. The stimulus materials included a news report regarding a crisis event (see ***Appendix A***) and the interaction record between a consumer and the online agent of the phone maker (see ***Appendix B***).

One fictitious news report about a crisis incident triggered by a swelling mobile phone battery was created for each vignette. Such an incident was chosen because it posed significant real-life risks that could escalate into a crisis for the manufacturer, such as device malfunction, battery leakage, fire and explosion hazards, leading to data loss, potential harm to users from chemical exposure or burns, and the possibility of injury or property damage. The news report indicated two actual battery swelling incidents had already occurred regarding the mobile phone maker Newcall's newly released batches of Malaysia-made products. However, the domestically produced phones in China were reported to be unaffected by this issue so far. The battery swelling issue posed serious safety risks and potential reputation damage for Newcall if not handled properly. Widespread battery failures could lead to product recalls, legal liability, loss of consumer trust, and regulatory scrutiny. Thus, it represented a crisis scenario where Newcall's crisis communication response would be crucial in protecting customers and mitigating harm to the company.

To ensure participants recognized the severity of the situation as a potential crisis, they were asked to evaluate two statements immediately after reading the news report: "What is the extent of the risk of a Newcall mobile phone user being harmed due to the battery swelling problem?" and "To what extent do you think Newcall will be involved in a crisis due to the battery-swelling incidents reported in Singapore?" These statements were rated on a 7-point scale from 1 ("not at all") to 7 ("totally agree").

Screenshots of the interactions between Newcall's agents (human or chatbot) and a concerned customer were then presented to participants. At the beginning of the conversation, both types of agents acknowledged the consumer's concern about the existence of a crisis event after self-introduction and then proposed that they could help check whether the consumer's phone was problematic upon inputting the batch number. The consumer was then informed that problematic phones would be registered by the agent for replacement with an updated version made in China, which had no such swelling problem. In both situations involving chatbots and human agents, the phone was verified as made in China and thus not problematic. However, the consumer still requested a replacement.

5.1.2. Measurement

The questionnaire consisted of two parts (see ***Appendix C***). The first part included manipulation check questions for agent type, message framing, and request handling. The second part included questions regarding the dependent and mediating variables. All measurements used a 7-point Likert scale.

Agent type. Using multiple-choice questions, the participants were asked whether the mobile phone company immediately sent a human or chatbot agent to communicate the crisis after reading the news report. The participants who gave a wrong answer were automatically excluded ($n = 4$). Using a two-item scale, they were then asked to rate the extent to which they believed the agent was a human/chatbot immediately after reading the WeChat dialogue record.

Response message framing. Using a nine-item measurement based on Page (2020), the participants were asked about the extent to which they agreed with the descriptions concerning the agent after reading the dialogue record.

Consumer's request-handling status. Using multiple-choice questions, the participants were asked about the agent's decision to agree or disagree with the consumer's replacement request. Those who gave the wrong answer were automatically excluded ($n = 3$). The participants were also asked about the extent to which they believed the agent had successfully processed the consumer's request.

Mediating variable. Perceived competence was measured using a five-item instrument on a 7-point Likert scale based on Cuddy et al. (2004).

Dependent variables. Responsibility attribution was measured based on Griffin et al.'s (1992) two-item scale. Satisfaction with the organization was measured using a three-item bipolar scale.

The reliability and validity results for the measurement model are shown in Table 2. The results implied a consistent and adequate level of

Table 2
Variable reliability and validity (Study 1).

		Minimal factor loading	Cronbach's alpha	Composite reliability	Average variance extracted
1	Instructing message	0.54	0.69	0.70	0.51
2	Adjusting message	0.75	0.90	0.93	0.68
3	Competence	0.76	0.85	0.88	0.65
4	Satisfaction	0.88	0.87	0.88	0.78
5	Responsibility attr	0.78	0.81	0.81	0.64

convergence validity across the entire model.

5.1.3. Participants and procedure

In our study, we used the Credamo (www.credamo.com) platform to recruit participants for our experiment. Credamo is a reputable Chinese data collection platform that offers a service analogous to the Qualtrics Online Sample service, providing access to a diverse and representative sample population from across China. The platform's widespread reach throughout various regions of the country ensures that our sample is not only large but also geographically and demographically diverse, thus enhancing the generalizability of our findings within the Chinese context. The platform distributed the survey to 410 individuals, of whom 11 participants were automatically excluded after failing the multiple-choice manipulation questions. The remaining 399 participants who gave valid responses (valid response rate = 97.3 %) were then recruited for financial compensation. Their average age was 31.28 (range = 18–58 years, *SD* = 7.97, 61 % female), and 89 % had completed higher education.

The participants were randomly assigned to one of eight vignettes. They were then asked to read a news report regarding the mobile phone battery swelling incident and answer multiple-choice manipulation check questions for agent type. Next, they were presented with a dialogue record between an agent from the manufacturer's official social media account and a mobile phone user. This was immediately followed by manipulation check questions for agent type, message framing, and request handling. Finally, the participants were requested to complete the items related to the dependent variables and the mediator. The entire process took approximately six minutes for each participant.

5.2. Results

Manipulation checks. After reading the news report, the participants showed a high degree of certainty (M = 5.71, SD = 1.13) that the manufacturer would undergo a crisis due to the battery swelling incident. They were likelier to believe that the agent was a human in the human agent condition (M = 5.57, SD = 1.82) than in the chatbot condition (M = 1.89, SD = 0.87) (t (397) = 23.66, $p < 0.001$). Similarly, the participants were likelier to believe the agent was a robot in the chatbot agent condition (M = 6.40, SD = 0.89) than in the human condition (M = 2.42, SD = 1.87) (t (397) = 27.24, $p < 0.001$). The participants perceived both instructing (M = 5.85, SD = 0.79) and adjusting messages (M = 5.97, SD = 0.52) as containing instructing information (t (397) = 1.78, p = 0.1). However, the participants in the instructing message condition perceived significantly fewer adjusting contents (M = 4.94, SD = 1.42) than those in the adjusting message condition (M = 5.73, SD = 0.80; t (397) = 6.83, $p < 0.001$). Furthermore, the participants expressed greater confidence that the agent had successfully processed the consumer's request in the condition in which the consumer's replacement request had been successfully registered (M = 6.33, SD = 1.08) than that which failed to be registered (M = 2.63, SD = 1.95) (t (397) = 23.45, $p < 0.001$). Thus, the results indicated that the manipulations of the agent type, response message framing, and consumer's request handling were successful.

Descriptive analysis. As shown in Table 3, the two dependent variables (responsibility attribution and satisfaction) were significantly correlated with perceived competence. However, the two dependent variables (responsibility attribution and satisfaction) were not significantly correlated.

Main effect. As shown in Table 4, while agent type had no significant main effect on the two dependent variables (responsibility attribution and satisfaction), it had a significant effect on perceived competence. In addition, chatbots were perceived as having a higher level of competence than humans. At the same time, message framing significantly influenced the two dependent variables and the mediator. Specifically, adjusting (vs. instructing) information was associated with lower responsibility attribution, higher stakeholder satisfaction, and higher perceived competence. However, request handling only had significant main effects on satisfaction and perceived competence but not on perceived responsibility. When the agent was able (vs. not able) to solve the request, participants ranked the agent higher in stakeholder satisfaction and higher in perceived competence.

Moderation effect. The only significant two-way interaction was between message type and request handling on the mediator (perceived competence). Specifically, when an agent failed to handle the crisis-related request, using adjusting messages (M = 3.82, SD = 0.63) was perceived as demonstrating higher competence than using instructing messages (M = 3.33, SD = 1.10; b = 0.49, SE = 0.10, CI [0.30 to 0.68], $p < 0.001$). In comparison, when an agent succeeded in handling a crisis-related request, there was no significant difference between adjusting (M = 4.01, SD = 0.43) and instructing (M = 3.90, SD =.59; b = 0.10, SE = 0.10, CI [−0.09 to 0.30], p = 0.30) messages. However, no other significant two-way interactions were found between other mediators and dependent variables through Model 1 of PROCESS (Bootstrap samples N = 5000, Hayes, 2022; see Table 5).

Model 3 of PROCESS (Hayes, 2022) was utilized to test the three-way interaction of online agent, message frame, and request handling on the mediator. Our results did not show any three-way interaction on the dependent variables; however, a significant three-way interaction among the three factors on perceived competence was found (b = 0.58, SE = 0.28; CI [0.03 to 1.12], p = 0.039).

In other words, when an agent failed to handle the crisis-related request, the chatbot agent using instructing messages (M = 3.53, SD = 0.92) was perceived to have higher competence than the human agent using the same messages (M = 3.13, SD = 1.05; b = 0.40, SE = 0.14, CI

Table 4
Descriptive statistics of ANOVA.

One-way ANOVA (F value)	Agent type	Message framing	Request handling
Perceived Responsibility	0.49, η^2 = 0.001 (M_{human} = 5.74, $M_{chatbot}$ = 5.66)	**5.06 *** , η^2 = 0.013 ($M_{instructing}$ = 5.83, $M_{adjusting}$ = 5.57)	1.82, η^2 = 0.005 (M_{fail} = 5.78, $M_{success}$ = 5.62)
Satisfaction	0.30, η^2 = 0.001 (M_{human} = 5.70, $M_{chatbot}$ = 5.77)	**12.01 * ***, η^2 = 0.036 ($M_{instructing}$ = 5.52, $M_{adjusting}$ = 5.95)	**76.45 * **** , η^2 = 0.168 (M_{fail} = 5.23, $M_{success}$ = 6.24)
Perceived Competence	**2.50 *** , η^2 = 0.013 (M_{human} = 3.68, $M_{chatbot}$ = 3.84)	**16.47 * **** , η^2 = 0.045 ($M_{instructing}$ = 3.62, $M_{adjusting}$ = 3.91)	**28.33 * **** , η^2 = 0.072 (M_{fail} = 3.57, $M_{success}$ = 3.95)

Note. Correlation significance level: *p < .05, * *p < .01, * **p < .001.

Table 3
Correlation matrix and descriptive statistics.

Measures	Perceived competence	Perceived responsibility	Satisfaction	*M*	*SD*
Perceived competence	/	−0.16 * *	0.71 * *	3.76	0.74
Perceived responsibility	−0.16 * *	/	−0.07	5.70	1.13
Satisfaction	0.71 * *	−0.07	/	5.73	1.26

Note. Correlation significance level: *p < 0.05, * *p < 0.01, * **p < 0.001.

Table 5
Index of the moderation effects on perceived competence/satisfaction/responsibility attribution.

Method	Outcome		
	M: Perceived competence	Y1: Satisfaction	Y2: Responsibility
XW: Agent type*Message frame (model 1)	−0.08(0.15) CI [−0.36 to 0.21]	−.07(.25) CI [−0.56 to 0.42]	0.22(0.22) CI [−0.22 to 0.67]
XZ: Agent type* Request handling (model 1)	−0.12(0.14) CI [−0.40 to 0.16]	−0.18(0.23) CI [−0.63 to 0.28]	0.38(0.22) CI [−0.06 to 0.83]
WZ: Message frame* Request handling (model 1)	−0.39(0.14) * * CI [−0.66 to −0.11]	−0.42(0.22) CI [−0.87 to 0.03]	−0.09(0.22) CI [−0.53 to 0.36]
XWZ: Agent* Message* Request handling (model 3)	**0.58(0.28) * CI [0.03 to 1.12]**	−0.02(0.45) CI [−0.92 TO 0.88]	0.25(0.45) CI [−0.63 to 1.13]

Note. Correlation significance level: *p < .05, * *p < .01, * **p < .001.

[0.13 to 0.67], $p = 0.004$). Furthermore, the chatbot agent using adjusting messages (M = 3.83, SD = 0.66) was perceived to have the same level of competence as the human agent using adjusting messages ($M = 3.80$, $SD = 0.62$; $b = 0.03$, $SE = 0.14$, CI [−0.24 to 0.31], $p = 0.81$). When the agent succeeded in handling the crisis-related request, the chatbot agent using adjusting messages ($M = 4.11$, $SD = 0.47$) showed no difference in competence perception compared to the human agent ($M = 3.90$, $SD = 0.37$; $b = 0.20$, $SE = 0.14$, CI [−0.07 to 0.48], $p = 0.15$). Similarly, when an agent was successful, there was no significant difference between the chatbot agent using instructing messages (M = 3.90, $SD = 0.62$) and the human agent ($M = 3.90$, $SD = 0.56$; $b = -0.01$, SE = 0.14, CI [−0.28 to 0.27], $p = 0.96$) in terms of perceived competence.

Moderated mediation. Model 11 of PROCESS (Bootstrap samples $N = 5000$; Hayes, 2022) was utilized to test the effect of chatbot use and artificial empathy expression in crisis communication. The results (see Table 6) revealed that perceived competence significantly mediated the three-way moderation effects of the online agent's (human vs. chatbot) instructing message framing (instructing vs. adjusting) in product return request situations (success vs. failure) on stakeholder satisfaction (b = 0.70, SE = 0.35, CI [0.04 to1.39]) and responsibility attribution (b = −0.13, SE = 0.07, CI [−0.30 to −0.01]).

Specifically, when an online agent failed to handle the stakeholder's request to return the product (see Fig. 2, upper part), the chatbot (M = 3.53, SD = 0.92) was perceived as having higher competence than the human agent (M = 3.13, SD = 1.05) when it showed instructing information, which resulted in higher satisfaction (b = 0.48, SE = 0.23, CI [0.03 to 0.98]) and less responsibility attribution ($b = -0.09$, $SE = 0.05$, CI [−0.22 to −0.01]). When an agent used adjusting information during the crisis, there was no significant difference between the chatbot (M = 3.83, SD = 0.66) and the human agent ($M = 3.80$, $SD = 0.62$) in terms of perceived competence, thereby having no effect on satisfaction (b = 0.04, SE = 0.16, CI [−0.27 to 0.34]) and responsibility attribution (b = −0.01, SE = 0.03, CI [−0.07 to 0.06]). Notably, the results suggested that the chatbot using adjusting information could perform equally well relative to its human counterpart in product return failure situations.

However, when an online agent succeeded in handling the stakeholder's request (see Fig. 2 lower part), the adjusting messages provided by the chatbot (M = 4.11, SD = 0.47) were associated with slightly higher competence than when those messages were delivered by a human agent ($M = 3.90$, $SD = 0.37$), thus resulting in higher satisfaction ($b = 0.24$, $SE = 0.11$, CI [0.03 to 0.45]) and less responsibility attribution ($b = -0.05$, $SE = 0.02$, CI [−0.10 to −0.01]). In comparison, there was no difference in satisfaction (b = −0.01, SE = 0.15, CI [−0.03 to 0.21]) and responsibility attribution ($b = 0.01$, $SE = 0.03$, CI [−0.06 to 0.06]) through the mediation of competence perception between the human ($M = 3.90$, $SD = 0.62$) and chatbot agents ($M = 3.90$, $SD = 0.56$) using instructing information.

Table 6
Index of the moderated moderated mediation effect through perceived competence.

Model 11	Outcome			
	M: Perceived competence		Y1: Satisfaction	Y2 : Responsibility
X: Agent type (chatbot=1, human=0)			Direct effect: c1′→ −0.12 (0.09)	c2′→0.04 (0.11)
W: Message frame (adjusting=1, instructing=0)				
Z: Request handling (success=1, failure=0)				
M: Perceived competence			**Indirect effect: b1→1.21 (0.06)*** **	b2→−0.23 (0.08)**
R^2	0.15		0.50	0.02
		Index	95 % bootstrap CI	
Moderated moderated mediation		**0.70 (0.35) * | −0.13 (0.07)***	**0.04 to 1.39**	**−0.31 to −0.01**
Conditional moderated mediation:				
By X: agent type (chatbot=1)	**(W=0, Z = 0)**	**0.48 (0.23) * | −0.09 (0.05)***	**0.03 to 0.98**	**−0.22 to −0.01**
	(W=0, Z = 1)	−0.01 (0.15) | 0.01 (0.03)	−0.30 to 0.21	−0.06 to 0.06
	(W=1, Z = 0)	0.04 (0.16) | −0.01 (0.03)	−0.27 to 0.34	−0.07 to 0.06
	(W=1, Z = 1)	**0.24 (0.11) * | −0.05 (0.02)***	**0.03 to 0.45**	**−0.10 to −0.01**

Note. Correlation significance level: *p < 0.05, **p < 0.01, ***p < 0.001.

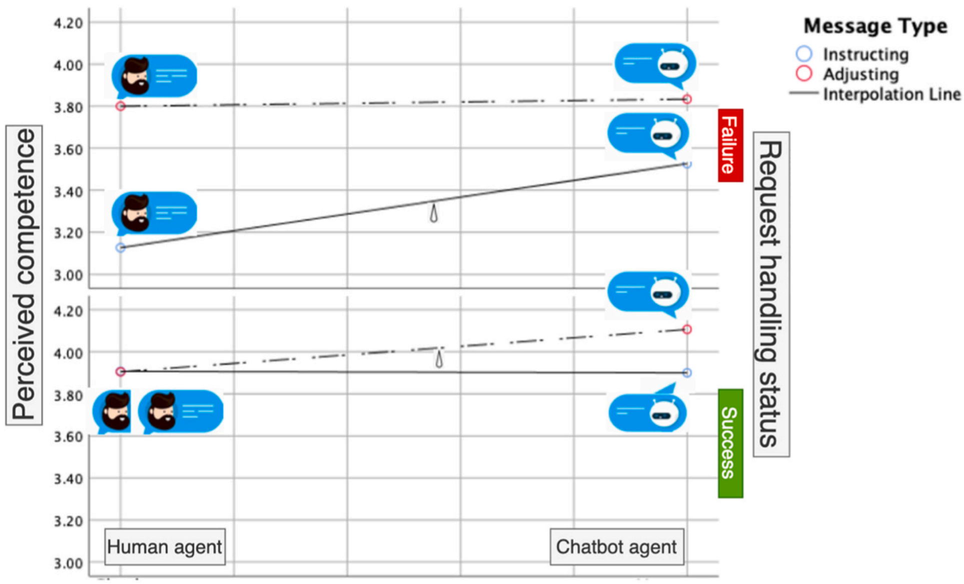

Fig. 2. The moderation of agent type by message framing on perceived competence at request handling failed and succeeded.

5.3. Discussion of study 1

When the stakeholder's crisis-related request is not resolved, using chatbots to communicate instructing information is more effective than humans in enhancing satisfaction and reducing responsibility attribution. However, there is no difference between the two types of agents for adjusting information (H1a). This finding complements the existing literature, which underscores the importance of direct and accurate communication in managing stakeholders' immediate responses during a crisis (Lee, 2018; Monroe et al., 2014) and highlights AI's advantages when it comes to processing functional and objective tasks (Logg et al., 2019; Ruan & Mezei, 2022; Wu et al., 2024).

Furthermore, the results revealed that chatbots are not advantageous compared to humans in providing adjusting information, especially in situations where the crisis has not been resolved. This is attributed to chatbots' inability to convey real empathy, provide psychological support, and establish relational connections (Yu et al., 2022), which are critical in addressing stakeholders' emotional needs and facilitating recovery in the face of an unresolved crisis (Claeys & Cauberghe, 2014; Schoofs et al., 2022). The findings echo prior studies on humans showing more tolerance of AI (vs. human) in service failure situations due to lower expectations of chatbots' adaptability (e.g., Garvey et al., 2023; Liu-Thompkins et al., 2022; Yalcin et al., 2022; Yu et al., 2022) and confirm the concept of machine heuristics (Kim & Sundar, 2019).

However, the current study's findings indicate that when an agent successfully handles a stakeholder's request, chatbots are more advantageous than humans in terms of adjusting information. Specifically, when the crisis-related request was resolved, chatbots (vs. humans) were able to enhance satisfaction and reduce responsibility attribution by giving adjusting information (H1b). The explanation is that when chatbots appear to "empathize" and tailor their responses to stakeholders' specific needs, users perceive chatbots as more competent than expected (Kim & Sundar, 2020). This is also echoed in the study of Castelo et al. (2023), who concluded that consumers initially reacted negatively to service bots, but their reactions improved when the bots consistently outperformed human service providers. Furthermore, this can also somehow explain why chatbots could enhance individuals' psychological resilience in the post-COVID-19 pandemic phase by providing empathic human–AI interaction (Jiang et al., 2022).

The results also show that perceived competence functions as a mediator of the three-way interaction of online agent, message frame, and request handling on satisfaction and responsibility attribution, such that more perceived competence leads to higher satisfaction and lower responsibility attribution (H2). Thus, the concept of "perceived competence" emerges as an important explanatory factor in the current study, mediating the relationship between the type of agent (chatbot versus human) and stakeholders' responses. This finding extends our understanding of the critical role played by competence perceptions in crisis communication with nonhuman entities, in accordance with existing research that highlights competence as a key driver of organizational trust and credibility (e.g., Claeys & Cauberghe, 2014; Schoofs & Claeys, 2021). To further establish the causal relationship between perceived competence and the dependent variable, we manipulated the mediator and examined its effect.

6. Study 2: the effect of chatbot competence

6.1. Methods

6.1.1. Study design and stimuli

We conducted an online experiment with a 2 (Competence of the chatbot: low vs. high) $\times$ 2 (crisis response framing: instructing vs. adjusting information) between-subjects design in the U.S.

The vignette used in the experiment is related to crisis communication during a fictitious event involving massive flight delays due to the extreme weather across the New York City area. The stimulus materials included a news report regarding a crisis event (see ***Appendix D***) and screenshots of the conversation between the online chatbot agent of the Federal Aviation Administration (FAA) and a passenger stuck at the airport, who was seeking help amid the flight delays. One fictitious news report about massive flight delays was created for each vignette: delays at the John F. Kennedy International Airport averaged over three hours. The FAA deployed an AI-based/pre-programmed customer service chatbot to assist passengers with their questions concerning connecting flights and departure times.

In the conversation screenshot (example in ***Appendix E***), a high-competence chatbot was deployed based on a powerful large language model like ChatGPT, while a low-competence chatbot used a pre-programmed script and matched passenger's requests to the most relevant questions with preset answers. Further, the low-competence chatbot used a more inflexible interaction approach. For example, it showed a list of questions from which the user can select. In contrast, the high-competence chatbot can understand the user's inquiries and respond flexibly. In addition, in the low-competence chatbot, we showed a conversation restart (matching error) in the screenshot, while this error was absent in the high-competence chatbot.

In both instructing and adjusting messages, the chatbot agent acknowledged the passenger's delay and then proposed that it could help to check airport status and delays upon inputting the airport name. Next, the chatbot instructed the passenger on how to deal with the connecting flight problem and explained why the exact departure time could not be provided due to the uncertain extreme weather conditions. Meanwhile, in the adjusting message framing condition, the chatbot also alleviated the passenger's anxiety and gave coping instructions to reduce distress due to prolonged delays. In both high- and low-competence level chatbot agent situations, the passenger's questions and reactions to the FAA were kept the same.

6.1.2. Measurement

The questionnaire consisted of two parts and had the same structure as in Study 1 (see ***Appendix F)***. All measurements used a 7-point Likert scale.

Chatbot agent competence level. Perceived competence was measured using a five-item instrument featuring a 7-point Likert scale based on Cuddy et al. (2004).

Response message framing. The participants were asked about the extent to which they agreed with the descriptions concerning the agent after reading the dialogue record through a nine-item measurement based on Page (2020).

Dependent variables. Attitudes toward the chatbot were measured based on Shin and Park's (2019) three-item scale. Satisfaction with the organization (FAA) was measured using a three-item bipolar scale.

The reliability and validity results for the measurement model are shown in Table 7. The results implied a consistent and adequate level of convergence validity across the entire model.

6.1.3. Participants and procedure

The MTurk platform distributed the survey to 226 individuals. Among them, 195 people have completed the survey and 189 valid responses were obtained after automatically excluding six participants who failed the multiple-choice manipulation questions (valid response rate = 83.6 %). Their average age was 30.53 (range = 21–55 years, SD = 5.08, 55 % female), and 94 % had completed higher education. The participants were randomly assigned to one of four vignettes after giving their consent. They were first asked to read a news report of the flight delay due to a series of storms passing through the New York City area and to answer multiple-choice attentional check questions. Then, they were presented with a dialogue record between the agent from the official APP of the FAA and a passenger stuck at the airport, immediately followed by manipulation check questions for agent competence level and message framing. Finally, the participants were requested to complete the remaining questions. The entire process took approximately four minutes.

6.2. Results

Manipulation checks. The participants perceived the high-competence chatbot agent to be more competent ($M = 6.07$, $SD = 0.59$) than the low-competence one ($M = 5.50$, $SD = 1.33$, $t\,(187) = 3.8$, $p < 0.001$). They also perceived both the instructing ($M = 5.86$, $SD = 0.71$) and adjusting messages ($M = 5.78$, $SD = 0.89$) as containing instructing information ($t\,(187) = .69$, $p = 0.53$). However, the participants in the instructing message condition perceived significantly fewer adjusting contents ($M = 5.55$, $SD = 1.29$) than those in the adjusting message condition ($M = 5.93$, $SD = 0.91$; $t\,(187) = 2.29$, $p < 0.05$). Thus, the results indicated that the manipulations of the chatbot agent's competence level and response message framing were successful.

Descriptive analysis. The two dependent variables (attitude toward the chatbot ($M = 5.88$, $SD = 0.98$) and satisfaction ($M = 5.80$, $SD = 0.99$) were both significantly correlated with each other ($r = 0.85$, $p < .001$).

Main effect. As shown in Table 8, message framing had no significant main effect on the two dependent variables (attitude toward the chatbot and satisfaction). This finding suggests that the crisis chatbots do not benefit from message framing. However, chatbot competence level had a significant effect on chatbot evaluation and organization satisfaction. In particular, the high-competence chatbot ($M = 5.99$, $SD = 0.65$) led to greater participant satisfaction with the FAA than the low-competence chatbot ($M = 5.58$, $SD = 1.24$, $p < 0.01$).

Moderation effect. There was no significant two-way interaction between message type and chatbot competence level in relation to either dependent variable (Table 5). This finding suggests that the effect of competence on the dependent variable does not depend on the message framing type.

Mediation effect. A mediation effect was found between chatbot competence level and satisfaction with the FAA through the passenger's attitude toward the FAA's chatbot via Model 4 of PROCESS (b =0.34, CI [0.11, 0.59], bootstrap samples $N = 5000$; Hayes, 2022). Specifically, the high-competence chatbot (compared with the low-competence chatbot) led to a more positive passenger of the agent ($b = 0.39$, $p = 0.005$, CI [0.12, 0.67]), which in turn, resulted in greater satisfaction with the organization ($b = 0.85$, $p < 0.001$, CI [0.77, 0.93]).

6.3. Discussion of Study 2

In Study 2, we investigated the effectiveness of two chatbots with varying competence levels during a public emergency crisis. The results reaffirm the significant role of AI competence in crisis communication, consistent with the findings of Study 1. The results also show that message framing has no influence on participants' satisfaction with the FAA or their attitude toward the chatbot. This finding aligns with Study 1, which highlights that message framing differences primarily affect users' perceptions of human agents rather than AI, particularly for failed requests. Moreover, the impact of competence appears to be independent of message framing, thereby implying a uniformly beneficial

Table 7
Variable reliability and validity (Study 2).

		Minimal factor loading	Cronbach's alpha	Composite reliability	Average variance extracted
1	Instructing message	0.69	0.76	0.76	0.52
2	Adjusting message	0.71	0.90	0.90	0.65
3	Chatbot competence	0.77	0.92	0.92	0.65
4	Satisfaction with FAA	0.74	0.84	0.84	0.64
5	Evaluation of chatbot	0.75	0.82	0.83	0.61

Table 8
Descriptive statistics of ANOVA.

	Satisfaction with the FAA	Attitude toward the chatbot
Message framing	$F(1,187) = 2.11$, $p = 0.15$, $\eta^2 = 0.011$ ($M_{ins} = 5.71$, $M_{adj} = 5.88$)	$F(1,187) = 1.95$, $p = 0.28$, $\eta^2 = 0.011$ ($M_{ins} = 5.82$, $M_{adj} = 5.94$)
Chatbot competence level	**$F(1,187) = 8.76$, $p = 0.003$ * *, $\eta^2 = 0.045$** ($M_{low} = 5.58$, $M_{high} = 5.99$)	**$F(1,187) = 8.45$, $p = 0.004$ * ** , $\eta^2 = 0.044$** ($M_{low} = 5.67$, $M_{high} = 6.07$)
Message*Chatbot competence	$F\,(1,185) = 0.56$, $p = 0.46$, $\eta^2 = 0.003$	$F(1,185) = 1.41$, $p = 0.24$, $\eta^2 = 0.008$

Note. Correlation significance level: *p < 0.05, * *p < 0.01, * **p < 0.001.

influence of perceived competence on enhancing both satisfaction and attitude. Furthermore, our analysis reveals that users' attitudes toward the chatbot could predict how its competence level impacts user satisfaction with the organization during a crisis. This finding suggests that users' perceptions of the chatbot's performance extend to their evaluations of the organization's crisis response.

This finding builds upon the work by Rese and Tränkner (2024), which underscores that chatbots that require less input, engage in shorter dialogues, use standard phrases, and have fewer conversation restarts can noticeably enhance users' perceptions of their conversational competence. As seen in our current study, the users' perceptions of chatbots' competence can further extend to their satisfaction with the organization's crisis response, regardless of the type of information (instructing or adjusting) conveyed.

Furthermore, the findings from Study 2, which examined a public emergency crisis scenario involving massive flight delays due to extreme weather, align with those from Study 1, which investigated an organizational crisis caused by mobile phone battery swelling. Both studies highlight the important role of AI competence in managing crisis communication effectively. The consistent results across different types of crises—public emergency and organizational—highlight the robustness of AI competence as a critical factor in shaping user satisfaction and attitudes. This finding also suggests that, regardless of the nature of the crisis, the ability of chatbots to competently handle stakeholder requests remains a key determinant of their effectiveness in crisis management. These insights reinforce the importance of investing in and developing high-competence AI systems for crisis communication in order to enhance the overall user experience and organizational reputation during challenging times.

While Study 1 and 2 highlighted the critical role of AI competence in crisis communication across different crisis scenarios, they primarily focused on a limited set of factors influencing user satisfaction. Recognizing that other antecedents likely contribute to positive perceptions of crisis handling, we conducted Study 3 as a follow-up exploratory study. This qualitative study aimed to broaden our understanding by exploring the public's general perceptions of and reactions to chatbot use during crises. By employing a questionnaire with open-ended questions, we sought to capture a wider range of user perspectives, identify common themes and patterns in their feedback, and uncover potential factors not addressed in the preceding quantitative investigations. This approach allowed us to complement and contextualize the findings of Studies 1 and 2, providing a more holistic view of chatbot effectiveness in crisis communication.

7. Study 3: a follow-up qualitative study

7.1. Methods

Study 3 adopted a qualitative design to explore user interactions with chatbots during crisis scenarios. To contextualize this inquiry, we present a detailed user scenario in which Sarah, impacted by a wildfire disaster, engages with a chatbot from the local authority for essential assistance. Her interactions include obtaining updates on the wildfire, locating shelters, finding emergency medical resources, and seeking emotional support guidance (see ***Appendix G*** for details about the user case). Subsequent to this scenario depiction, the study participants were immediately asked to articulate their perspectives on chatbot use in crises through a structured questionnaire featuring open-ended questions to garner rich, detailed responses. They were prompted to put themselves in Sarah's shoes and imagine how they might react to and value the use of chatbots for assistance during a similar crisis. Finally, the participants were encouraged to share their views on the strengths and potential limitations of deploying chatbots in such situations. Their responses were expected to be concise, around 100 words per question. The participant pool comprised 121 individuals with an average age of 41.2 years (SD = 12.8), including 45 males and 76 females, with three nondisclosures. Recruitment was conducted via Prolific (www.prolific.com) without specific exclusions. The platform distributed the survey to 171 individuals (response rate = 70.8 %).

The qualitative feedback underwent topic modeling analysis, adhering to the protocols recommended by Yu (2023) for leveraging LLMs in analyzing short textual data. This approach is particularly effective for the topic modeling of short textual responses. Three LLMs, namely, GPT 4 Turbo, Gemini 1.0 Ultra, and Claude Sonnet, were deployed to ascertain dominant themes, ensuring analytical robustness through cross-validation. Discrepancies in topic granularity were addressed by comparing outcomes, fine-tuning categorizations, and integrating related themes for coherence (see ***Appendix H*** for the detailed procedure).

The consistency in thematic extraction was notable, particularly between GPT 4 Turbo and Claude Sonnet, thus demonstrating parallel findings in topic quantity and nature. In comparison, Gemini 1.0 Ultra exhibited fewer themes—a variance explored and shown in the appendix for transparency and comprehensive understanding.

7.2. Results

The analysis of users' reactions to and feelings about crisis chatbots revealed mixed opinions. The majority of users acknowledged positive attributes in chatbots, appreciating their capacity to provide rapid assistance and access to crucial information when most needed. These individuals valued the chatbots' ability to offer immediate support and direct users toward the necessary resources. These findings are consistent with those of Study 2. Furthermore, the users valued highly competent chatbots in terms of resolving crisis-related matters. Conversely, concerns were voiced about potential negative aspects of deploying chatbots during critical moments. In particular, the skepticism revolved around chatbots' lack of emotional empathy and potential misunderstandings of users' complex situations, consistent with the findings of the two previous studies. Thus, using chatbots to give adjusting information may not be as advantageous as expected. Amid these contrasting views, the users proposed enhancements for chatbot functionalities. Their recommendations included augmenting chatbots' emotional intelligence, ensuring robust data privacy, and establishing clear guidelines for human intervention. To elaborate, we summarized the perceived strengths, concerns, and suggestions for crisis chatbots.

7.2.1. Perceived strengths of using chatbots for crisis handling

The respondents believed that one of the most significant advantages was chatbots' **immediate accessibility**, as mentioned in the response, *"I imagine the bots are available 24/7 unlike people who might be affected by the crisis."* Chatbots operate around the clock, ensuring that crucial information and assistance are accessible at any time without delay, which is particularly vital during emergencies when time-sensitive guidance is needed. Another strength emphasized is **scalability**. One respondent noted that chatbots *"can be available to a lot of people at the same time and provide useful information."* This scalability allows chatbots to engage with multiple users simultaneously, distributing essential information efficiently to a large audience without being overwhelmed. As one response stated, *"The biggest strength is that the chatbot can send and give information to many people at once and can be continuously updated with new information that people can access rapidly."*

Consistency is also a crucial strength highlighted in the responses. As one respondent mentioned, *"The same information can be given to everyone, and this info can be easily kept updated."* Chatbots provide consistency, ensuring that the information shared is standardized and reliable, which is crucial for maintaining order and trust during a crisis.

The respondents also recognized the **resource efficiency** offered by chatbots. As one response stated, *"It frees up other resources—people—to places they are needed."* By handling routine inquiries, chatbots can free up human resources to focus on more complex or critical tasks, thereby optimizing overall response efforts during a crisis. Chatbots' ability to provide **rapid information updates** is another strength highlighted in the responses. As one response noted, *"They can be updated and can get to many different people at one time."* Chatbots can be quickly updated with the latest information, ensuring that the public receives timely and accurate guidance as the given situation evolves.

Furthermore, the respondents emphasized the **emotional neutrality** of chatbots. As one response mentioned, *"There are no stress factors or human emotions that can get in the way of [chatbots'] ability to assist quickly and efficiently."* Unlike humans, chatbots are unaffected by stress or emotions, ensuring that they deliver calm and rational advice, which can be reassuring to users in distress.

Finally, the respondents recognized chatbots' ability to **direct resource connections**. As one response stated, *"They can accurately assist in emergency situations. They can access data and instruct on nearby facilities."* Chatbots can guide users to appropriate resources or escalate issues to human operators when necessary, thus streamlining the support process.

7.2.2. Concerns

Apart from the perceived advantages, the respondents also believed that the use of chatbots in crisis situations presented several challenges that must be addressed. One of the primary issues highlighted is the **impersonal nature** of chatbots, which is crucial in providing comfort and support to individuals facing distress during a crisis. As one respondent mentioned, *"They aren't human and therefore lack emotional capacity."* Moreover, the dependency on technology is another significant concern. The respondents pointed out that *"Without electricity, you can't access them"* and that *"Internet issues"* could hinder the effectiveness of chatbots in emergency situations. In a crisis, infrastructure damage may occur, rendering chatbots inaccessible when they are needed the most. The **accuracy and timeliness** of the information provided by chatbots are also questioned. In particular, the respondents expressed worries such as, *"How accurate is the information that is being presented? Where is it getting this data from?"* and *"Not real-time information,* i.e.*, who is updating the chat bots to the situation and how frequently?"* Inaccurate or outdated information could have severe consequences in rapidly evolving crisis scenarios.

Furthermore, **technical limitations and failures** of chatbots are another area of concern. As one respondent stated, *"A malfunction. Incorrect information. Still the uncertainty of a chat bot; if we don't use it often, it would feel like a novel approach, which, given the situation, would be worrying."* System crashes or technical glitches could leave individuals without support during critical times.

The **lack of personalization** in chatbot responses is also seen as a drawback. For example, the respondents mentioned that *"Responses are likely to be generic rather than specific to a unique individual and their specific circumstances"* and *"The chatbot cannot comfort people in a time of crisis and does not have the ability to empathize with them. Its usefulness is generic."* Chatbots may struggle to address specific individual needs or provide tailored guidance in complex situations.

Finally, **security and privacy concerns** were raised, particularly regarding the sharing of sensitive information during crises. The respondents expressed concerns about *"whether it could be hacked or if the system could fail"* and *"Doubts may arise regarding the authenticity of content provided."* The potential for data breaches or misuse of personal information shared with chatbots could deter individuals from relying on them during emergency situations.

7.2.3. Discussion

This qualitative study confirmed that, while the unique features of chatbots can offer significant benefits, they also present certain limitations that could potentially undermine their effectiveness in crisis situations. The discussion sheds light on the double-edged nature of chatbot functionalities, which can be seen as having both positive and negative implications, depending on the context and implementation.

One primary feature of chatbots is their ability to provide consistent and standardized responses (Yu et al., 2022). This feature can be highly advantageous in crisis situations, where delivering accurate and reliable information quickly is crucial. However, this same consistency may limit the chatbot's ability to provide flexible, personalized, and context-specific responses, which are often necessary to effectively address the unique aspects of an individual's crisis situation.

Another factor to consider is the chatbot's availability and quick response time (Ashfaq et al., 2020). These features ensure that individuals in crisis can receive immediate support anytime, which is especially important when human counselors are unavailable. Nonetheless, this immediate response capability does not equate to the depth and empathy of human interaction, potentially making chatbots seem impersonal or inadequate in meeting humans' complex emotional needs (Meng & Dai, 2021).

Future research should explore strategies for mitigating the limitations of chatbots while maximizing their strengths. For instance, investigating how AI can be designed to provide more personalized and context-aware responses, potentially through incorporating user profiles or integrating with external data sources, is crucial. Additionally, exploring the ethical implications of chatbot use in crisis situations, particularly concerning data privacy and the potential for bias, is paramount. Further research could also examine the optimal balance between automated support and human intervention, identifying instances where human escalation is necessary to provide comprehensive and empathetic care.

8. Discussion

This study embarked on an in-depth analysis comparing chatbots to human agents in delivering instructing versus adjusting information, particularly under the pressures of a crisis, while taking into account whether or not the stakeholder's crisis-related demands were successfully resolved. The study's findings indicate that chatbots (vs. humans) are particularly effective in delivering instructing (vs. adjusting) information during unresolved crises. Our follow-up qualitative survey also confirms this finding, as users perceive that the value of crisis chatbots lies in scenarios demanding immediate, clear, and unambiguous instructions, in which case the consistency and reliability of chatbots can be highly advantageous. The study findings also revealed the prominent perceived values of and concerns in using chatbots for crisis communication due to chatbots' inherent attributes. Individuals valued the chatbots' ability to offer immediate support and direct users toward the necessary resources. Conversely, concerns revolved around chatbots' lack of emotional empathy and potential misunderstandings of users' complex situations.

Collectively, these insights advocate a balanced and strategic integration of chatbots and human agents in crisis communication frameworks. In particular, organizations can design and implement more nuanced and effective communication strategies by simultaneously capitalizing on chatbots' strengths in delivering direct, instructing information during the acute phases of a crisis and harnessing human agents' ability to provide empathetic, adjusting information. Such an approach not only aligns with the theoretical underpinnings found in crisis communication literature, but also resonates with the practical considerations of organizations aiming to maintain stakeholder trust and mitigate reputational damage during crises.

8.1. Theoretical contributions

First, the present study contributes to the understanding of using chatbots to disseminate information during organizational crises through the lens of machine heuristics. While previous studies have focused on consumer experiences with service request rejections or complaints (e.g., Crolic et al., 2022; Garvey et al., 2023; Yu et al., 2022), the current study investigates consumer responses to chatbot agents' handling of crisis-related requests. Crises create complex emotional states, such as anxiety, beyond anger (Jin & Pang, 2010). The findings of this study indicate that, in unresolved crisis contexts, chatbots can be perceived as advantageous over humans in communicating crisis-related information. This is in line with prior studies suggesting that people typically have lower expectations of chatbots' ability to handle unexpected situations (Crolic et al., 2022; Yu et al., 2022) because of chatbots' unique inherent nature (Sundar & Kim, 2019). This also confirmed the "moral crumple zone" concept, in which AI agents can absorb responsibility (Hohenstein & Jung, 2020). Thus, using chatbots to deliver bad news can be rather effective (Garvey et al., 2023; Mozafari et al., 2022; Yu et al., 2022).

Second, the study's findings offer valuable insights into SCCT in the AI era. Prior studies have mostly applied SCCTs in their investigations of regular media, such as television, newspapers, and social media (Coombs, 2017; Schultz et al., 2011). The present study extended SCCT to a nonhuman touchpoint and investigated the role of base crisis response information (i.e., instructing/adjusting) when using chatbots for crisis communication. Regardless of their ability to handle stakeholders' requests, chatbots are advantageous over humans in disseminating information during a crisis, although this depends on the message type. In particular, when chatbots fail to handle requests, they are still considered superior to humans in delivering instructing information, given that people have lower expectations of chatbots' flexibility during a crisis due to their lack of human-like intelligence and ability to adapt to unexpected situations (Yu et al., 2022). However, when they can successfully satisfy users' crisis-related requests, chatbots are considered more effective than humans in providing adjusting information because users may perceive the chatbots as more competent than expected when chatbots constantly show empathy and address their demands (Castelo et al., 2023; Kang and Kim, 2022). This study thus reveals that SCCT should leverage both chatbot and human efforts to achieve effective crisis communication in the AI era.

Third, this study extends the research stream on artificial empathy for chatbots. Prior research has shown mixed findings on whether chatbots should express empathy. Liu and Sundar (2018) revealed that chatbots expressing sympathy and empathy are favored over the unemotional provision of advice, especially by those who are initially skeptical about machines possessing social cognitive capabilities. Jiang et al. (2022) investigated empathy in communication between chatbots and humans during the COVID-19 pandemic and found that when processing information obtained from AI and collaborative interactions with the AI chatbot, various dimensions of mediated empathy unexpectedly served as pathways to resilience, thus contributing to the enhancement of users' well-being. In this study, we have shown that the phenomenon of chatbots showing empathy conveyed by adjusting messages was linked to reduced responsibility attribution, increased stakeholder satisfaction, and heightened perceived competence, especially when requests were resolved in the crisis. However, when requests during a crisis are unresolved, the adjusting information provided by chatbots is not more advantageous than that provided by humans. This finding aligns with that of Yu et al. (2022), who suggested that artificial empathy by chatbots may not be effective for unresolved service requests. This observation is further supported by research indicating that the use of empathic features in chatbots can lead to negative outcomes, particularly when dealing with angry customers (Crolic et al., 2022; Liu-Thompkins et al., 2022).

Finally, the present study extends previous research, which emphasized enhancing the competence of human spokespersons such as CEOs in image repair and reputation management (e.g., Claeys & Cauberghe, 2014; Schoofs & Claeys, 2021), to include non-human entities such as chatbots. As anticipated, our findings suggest that perceived competence plays a mediating role in the relationship between chatbots/human agents and stakeholders' satisfaction and responsibility attribution in crisis communication. Furthermore, the results reveal that the competence of chatbots utilized by an organization can significantly impact stakeholders' perceptions of the organization during a crisis. This highlights the importance of effectively leveraging non-human entities in crisis management strategies.

8.2. Managerial implications

The study's findings provide substantial managerial implications for organizations and suggest several strategic approaches to leveraging chatbots and human agents in crisis communication.

First, organizations should dynamically evaluate the information that is available and ready to share with stakeholders before deploying chatbots to disseminate information. Our findings indicate that when a stakeholder's crisis-related request cannot be resolved, using chatbots to communicate instructing information is more effective than using human agents. This pre-evaluation ensures that chatbots can personalize their communication strategies effectively by estimating whether a stakeholder's crisis-related request can be fulfilled or must be rejected. For instance, during the initial stages of a crisis, certain data or decisions might still be under investigation or verification, making them unsuitable for immediate release. By identifying such information in advance, organizations can instruct chatbots to communicate transparently about ongoing efforts to clarify these details while avoiding the provision of incomplete or incorrect responses. This preemptive strategy not only preserves the credibility of the organization but also helps manage stakeholder expectations more effectively.

Moreover, chatbots can be equipped with decision-making protocols that estimate whether a stakeholder's request can be fulfilled immediately or should be deferred. When a request cannot be met right away, chatbots can provide a rationale for the delay and offer alternative sources of information or support. This personalized approach fosters a sense of understanding and patience among stakeholders, thus reducing frustration while providing a personal touch. For example, in the event of a natural disaster, a utility company might need to prioritize the release of safety instructions and outage updates while delaying specific details about restoration timelines until more accurate information is available. In this case, the company can ensure that its chatbots deliver timely and pertinent messages, keeping stakeholders informed and engaged without overpromising or underdelivering.

Second, given chatbots' effectiveness in delivering instructing information during unresolved crises, chatbots can assist in prioritizing and disseminating the most critical and actionable information, such as safety instructions, emergency procedures, and immediate steps that must be taken by stakeholders. By focusing on the immediate needs of stakeholders and providing clear guidance, chatbots can help mitigate the impact of the crisis, while more complex details are still being verified. This can be particularly beneficial in the acute phases of a crisis, during which swift and accurate communication is crucial in managing stakeholders' immediate responses. To address concerns about potential misunderstandings of complex and uncertain situations immediately after a crisis, chatbots should be designed to guide users through a structured interaction process designed to restore order during a chaotic crisis situation. This action involves creating an intuitive and user-friendly interface that can lead stakeholders through a step-by-step breakdown of necessary procedures, clarify any ambiguities, and provide tailored responses based on users' specific queries and needs. This initiative can help ensure that critical information is not missed and that user queries are accurately understood and addressed. For example, in the event of a cybersecurity breach, a technology firm might use chatbots to activate a structured interaction process to inform users about immediate protective measures they should take, such as changing passwords and monitoring account activity. While the firm may not yet have full details regarding the extent of the breach, the chatbot can keep users updated on the investigation's progress and provide assurance that the firm is actively working to resolve the issue. Similarly, during a public health crisis, a healthcare provider could deploy chatbots to communicate essential information about safety measures, symptoms to watch for, and where to seek medical help. Thus, even if specific details about the outbreak are still being confirmed, the chatbot can already step in and play a crucial role in guiding the public through the initial response phase.

Third, our findings indicate that when crisis-related requests during a crisis are unresolved, chatbots' adjusting information is not more advantageous than that provided by humans. Our qualitative study further highlights chatbots' inability to convey empathy and establish relational connections, which are critical in addressing stakeholders' emotional needs during unresolved crises. Therefore, human agents remain indispensable, especially in providing adjusting information that facilitates recovery and reassures stakeholders in such cases. Organizations can implement an integrated approach by training chatbots to handle routine queries and disseminate essential information quickly while reserving human agents for more complex and emotionally charged interactions. In the immediate aftermath of a crisis, stakeholders often experience heightened anxiety and uncertainty, which can lead to increased scrutiny and criticism of the organization involved. Thus, in unresolved crisis scenarios, such as the acute phase of a crisis, it is particularly important to monitor stakeholders' emotions during their interactions with chatbots. When stakeholders exhibit high levels of negative emotions and distress, it may be an indication that they require more empathetic responses than a chatbot can provide. In such cases, chatbots could be programmed to detect these cues and seamlessly transfer the conversation to a human agent. This step ensures that stakeholders receive the emotional support and personalized care they need during a challenging time. For instance, a retail company might use chatbots to issue immediate recall instructions for a defective product, ensuring that all customers receive information quickly and accurately. Subsequently, human customer service representatives can handle individual concerns, provide emotional reassurance, and offer more personalized and empathetic responses to customers affected by the product defect.

Furthermore, this study underscores the critical role of perceived competence in mediating stakeholders' responses to crisis communication. In particular, our findings indicate that stakeholders favor LLM-based chatbots over preprogrammed chatbots, as the former are perceived to be more competent in handling crisis-related queries. This perception not only enhances stakeholders' satisfaction with the bots but also extends to the organization using these bots. This preference highlights the importance of leveraging advanced generative AI technologies to enhance stakeholders' perceptions of chatbots' competence for crisis management. Therefore, organizations should design chatbot interactions to enhance perceived competence. For instance, the implementation of chatbots in crisis communication should be complemented with robust monitoring and analytics capabilities. This allows organizations to track user interactions, identify common concerns or misconceptions, and adjust their chatbots' responses accordingly. Such data-driven insights can provide valuable feedback for the continuous improvement of the chatbot system, ensuring that it remains responsive and effective in managing emerging crises. Moreover, integrating a feedback loop in which stakeholders can rate their interactions with chatbots offers a valuable source of qualitative data that may enhance chatbots' perceived competence. By capturing stakeholders' immediate reactions and assessments, organizations can gather insights into the effectiveness of their chatbots' responses, identify areas for improvement, and discover recurring issues or gaps in information.

8.3. Limitations and future research directions

This study has some limitations that, in turn, point to avenues for future research. First, there are potential variables concerning the characteristics of organizations, crisis events, and stakeholders to be tested to determine the effectiveness of AI-mediated crisis communication. For example, an organization's precrisis reputation would largely influence stakeholders' trust in its chatbots (Coombs & Holladay, 2006). On the one hand, if an organization has a favorable reputation in terms of transparency, responsiveness, and honesty, stakeholders may be likelier to trust chatbots as crisis communication tools. On the other hand, if an organization has a history of being unresponsive or dishonest, stakeholders may be less likely to trust chatbots or may even be skeptical of their intentions (Coombs, 2004). Other variables that could potentially influence the effectiveness of AI-mediated crisis communication include whether the severity and type of crisis, size, and demographics of the stakeholder population.

Furthermore, it would be interesting to explore how these characteristics interact with chatbots' specific characteristics, such as communication tone and style (Whang et al., 2022), and the perceived level of expertise. For example, a chatbot that is perceived as humorous might be less effective for an organization that already has a negative reputation (Claeys and Cauberghe, 2015) or is involved in a higher-level severity crisis (Xiao et al., 2018). These factors should be carefully considered and tested in future studies investigating the effectiveness of AI-mediated crisis communication.

Second, in the current paper, we combined chatbots' capabilities as perceived competence, thereby obtaining a more comprehensive dimension. However, it is worth noting that different aspects of these capabilities significantly influence the process of AI-mediated communication and users' attitudes toward chatbots. For example, factors such as conversational tone (Jiang et al., 2022), communication delay (Cheng et al., 2022), emotions (Coombs and Holladay, 2005;Tsai et al., 2021), and tailored responses (Jiang et al., 2023), each have distinct impacts on communication quality.

More importantly, although perceived competence serves as a significant mediator in explaining the differences between humans and robots, many other factors could influence the affordances of AI chatbots in supporting crisis communications. Therefore, future research should consider revealing the underlying mechanisms of AI-mediated communication from various perspectives. Furthermore, it is noteworthy to investigate other mediating variables, such as perceived warmth (Yu and Zhao, 2024), social presence (Munnukka et al., 2022; Kang & Kim, 2022; Kim et al., 2022), and transparency (Liu, 2021; Lopez & Garza, 2023; Yu & Li, 2022).

9. Conclusion

This research offers novel insights into the potential of AI-powered chatbots in crisis communication by comparing their effectiveness with human agents across various conditions of crisis management. The findings suggest that chatbots, particularly when disseminating instructing information in unresolved crisis scenarios, can enhance stakeholder satisfaction and reduce responsibility attribution toward the organization, which can be attributed to the heightened perception of competence. In addition, the study reveals the context-dependent superiority of chatbots over humans, especially when chatbots successfully address crisis-related requests by adjusting information, thus boosting perceived competence and stakeholder satisfaction.

Importantly, this research sheds light on the critical role of perceived competence as a mediating factor that influences both stakeholder satisfaction and responsibility attribution. By emphasizing the importance of strategic message framing and enhancing chatbot competence to optimize stakeholder responses, these findings not only extend the theoretical understanding of AI-mediated crisis communication but also offer pragmatic guidelines for organizations contemplating the integration of chatbots into their crisis communication strategies.

CRediT authorship contribution statement

Yi Xiao: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Shubin Yu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no competing interests in relation to this paper. The authors have received no funding or support from any organization or institution that may have influenced the design, execution, analysis, or interpretation of this study. The authors have full control over all primary data and agree to allow the journal to review their data if requested.

Data Availability

The data for this article will be shared on reasonable request to the corresponding author.

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Appendix A. : News report of Study 1

Newcall's new mobile phone "Newcall-3" batteries are swelling up.

The product, which is produced by Newcall's manufacturing partner in Malaysia, has been reported to have possible battery bulge issues. According to The Straits Times in Singapore, the Singaporean telecommunications authority has received two complaints regarding battery bulges. No similar issues have been reported in domestically produced Newcall mobile phones.

It is understood that Newcall mobile phones produced in Malaysia are mainly sold in Southeast Asia, with a small portion of products returning to the domestic market. Mobile phones on the domestic market are mostly produced domestically. In response to the battery issue, Newcall Company has set up an intelligent chatbot/ human customer service representative on its official social media account to handle user queries.

Appendix B. : The scripts of Study 1

To manipulate the online agent type, in the chatbot agent situation, the news report indicated the phone maker used a chatbot on social media (i.e., *WeChat*) to communicate all the questions regarding the crisis from stakeholders. Moreover, the agent was also self-introduced as an AI chatbot service representative at the beginning of the dialogue. In the human agent situation, the news report indicated the phone maker used a human service representative to communicate all the queries.

Regarding the response message framing, in instructing the message, the agent asked the consumer if the phone had overheating problems when using it, and then the know-how to protect themselves from these unsafe situations was provided. Besides that, the consumers also acknowledged that for the sake of the user's safety, an investigation concerning the swell-up problem has already been activated by the manufacturer. In the adjusting message, the agent asked the consumers if they were suffering from worry and anxiety that the phone would have overheating problems when using it. In addition, the agent offered a coping strategy to reduce consumers' anxiety and worry by providing the know-how to protect themselves from these unsafe situations (same as instructing information). Besides that, the consumers also acknowledged that the manufacturer fully understood how they felt.

Regarding the consumer's request handling, in the request handling success situation, the replacement request was successfully registered, and the consumer could replace the phone at any chain store offline. However, the message was framed as 'for the sake of *user's safety*', we agree on the replacement request' in instructing condition. While in the adjusting situation, the message was framed as 'for the sake of *user's reassurance*', we agreed on the replacement request. In the request handling failure situation, the agent refused the replacement request. The message was framed as 'the mobile phone is made in China, which is not belonging to the problematic batches of products. We confirm the phone is in a safe condition and no need for replacement' in instructing message; and '*we understand your worries, however,* the mobile phone is made in China, which is not belonging to the problematic batches of products in the adjusting message.

Appendix C. : Measurements of Study 1

a. Agent type.

- The technical specialist agent in the dialogue is a human.
- The technical specialist agent in the dialogue is a robot.

b. Response message framing:
Instructing Information.

- New-call informed people who could be hurt.
- New-call helped people know to get to safety when using their cell phones.
- New-call told people how to protect themselves.
- New-call gave information to protect people who could be harmed.

Adjusting Information.

- New-call gave information to ease suffering after the event.
- New-call provided emotional support for people involved.
- New-call helped people emotionally.
- New-call demonstrated its commitment to helping people involved.
- New-call expressed sympathy for people might be harmed in the event.

Consumer's request handling

- To what extent do you believe the agent has successfully processed the consumer's request for the cell phone replacement?
  Perceived competence of the agent
- To what extent do you agree the online agent is skilful/ efficient/ competent/ intelligent/flexible?

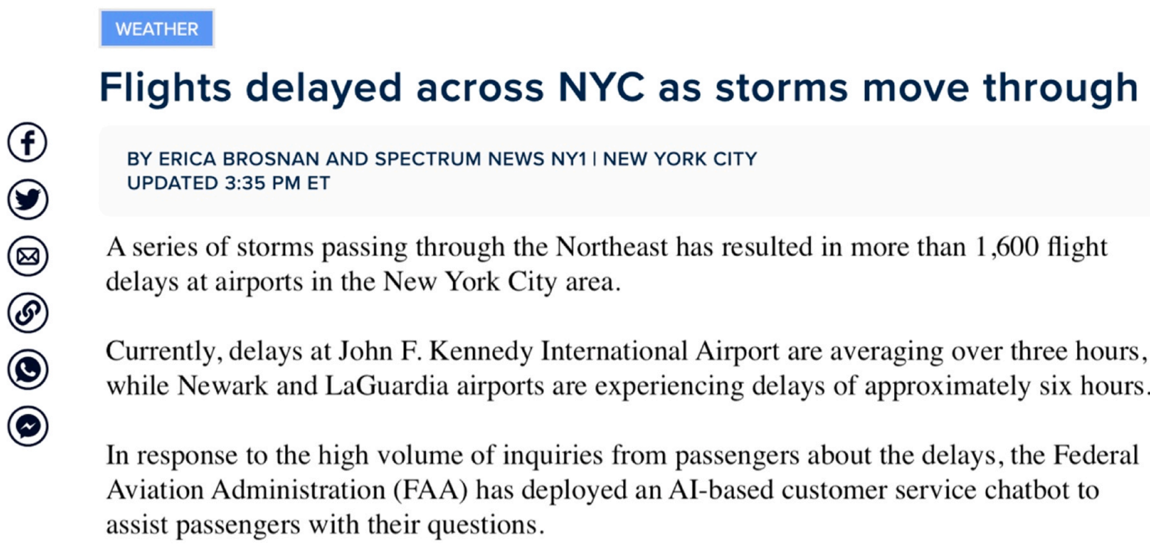
c. Responsibility attribution

- How responsible was the manufacturer with respect to the crisis?
- To what extent do you blame the manufacturer for the crisis?

d. Satisfaction with the organization.
To what extent do you satisfy with the reaction from the manufacturer on a seven-point Likert bipolar scale?

- Completely unacceptable to 7-completely acceptable
- Extremely bad to 7- extremely good
- Completely unsatisfied to 7-completely satisfied.

Appendix D. News report of Study 2



WEATHER

Flights delayed across NYC as storms move through

BY ERICA BROSNAN AND SPECTRUM NEWS NY1 | NEW YORK CITY
UPDATED 3:35 PM ET

A series of storms passing through the Northeast has resulted in more than 1,600 flight delays at airports in the New York City area.

Currently, delays at John F. Kennedy International Airport are averaging over three hours, while Newark and LaGuardia airports are experiencing delays of approximately six hours.

In response to the high volume of inquiries from passengers about the delays, the Federal Aviation Administration (FAA) has deployed an AI-based customer service chatbot to assist passengers with their questions.

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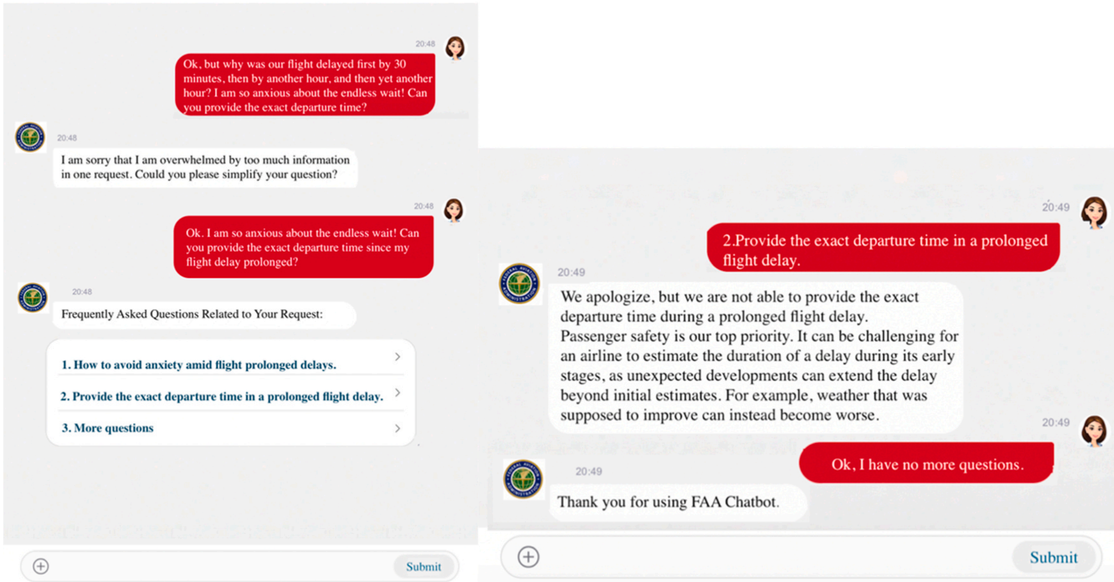
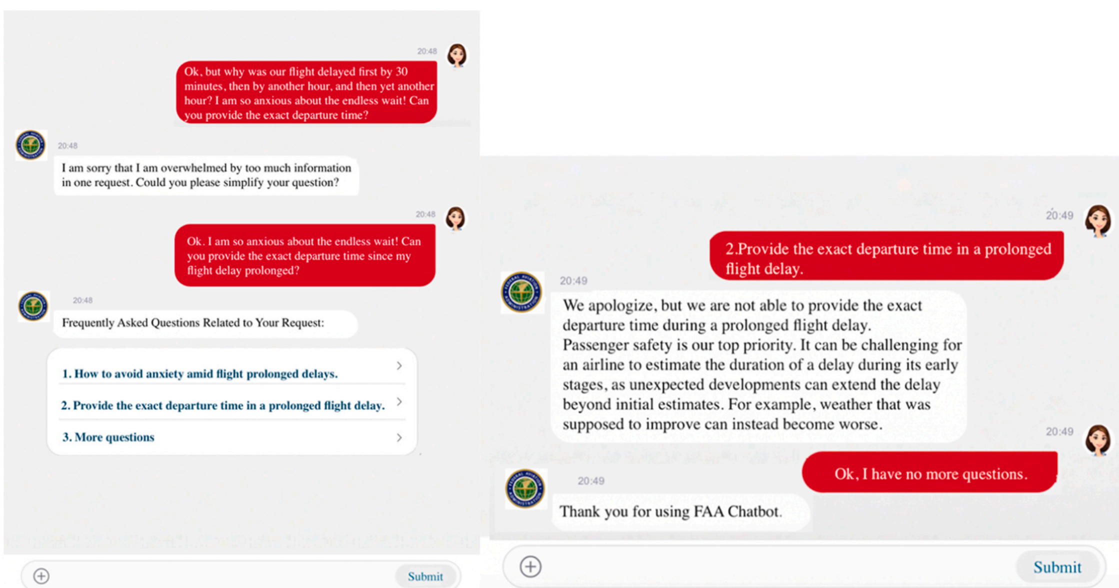
Appendix E. The conversation scripts of Study 2

A. **Low competence * instructing information**





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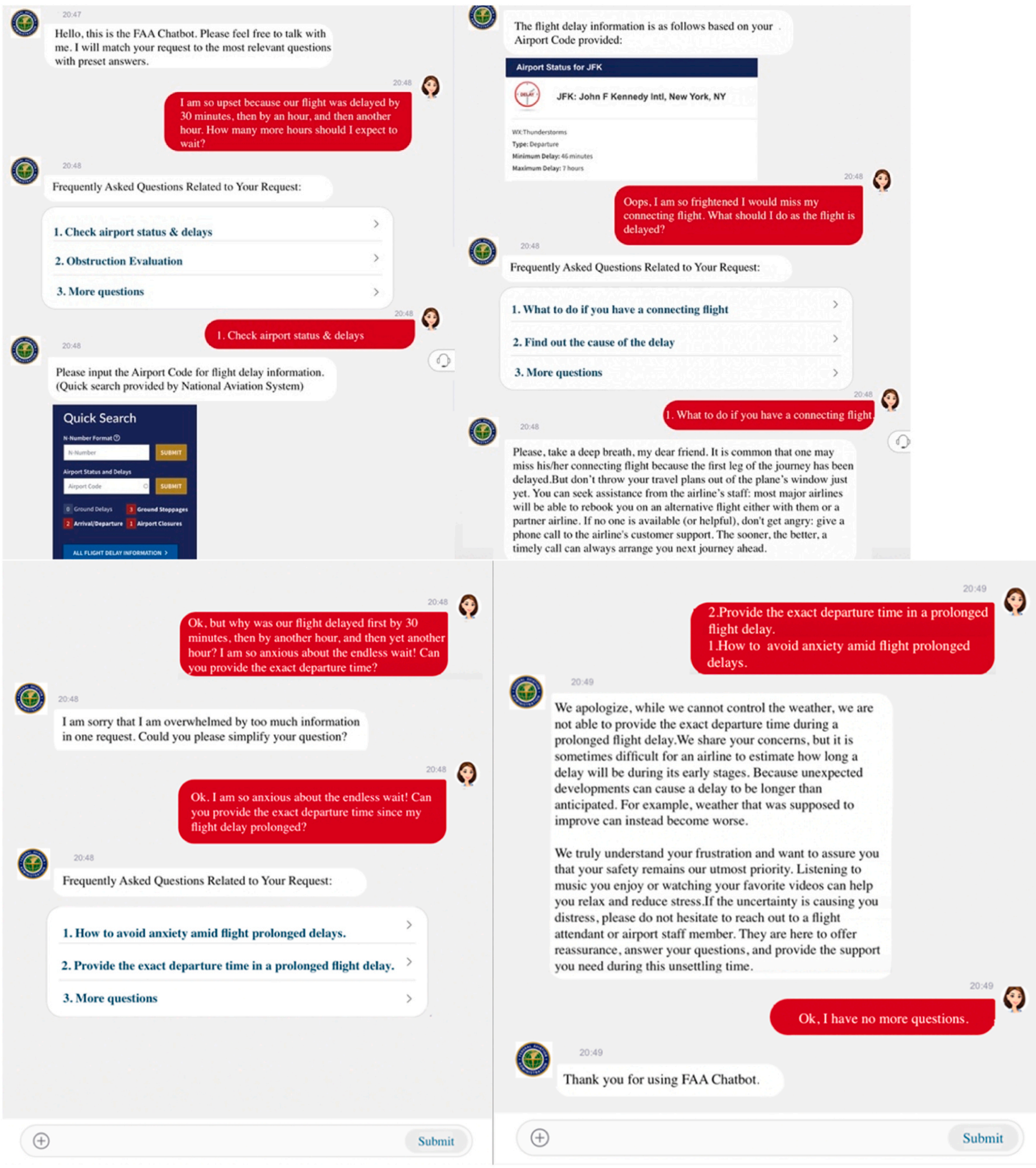
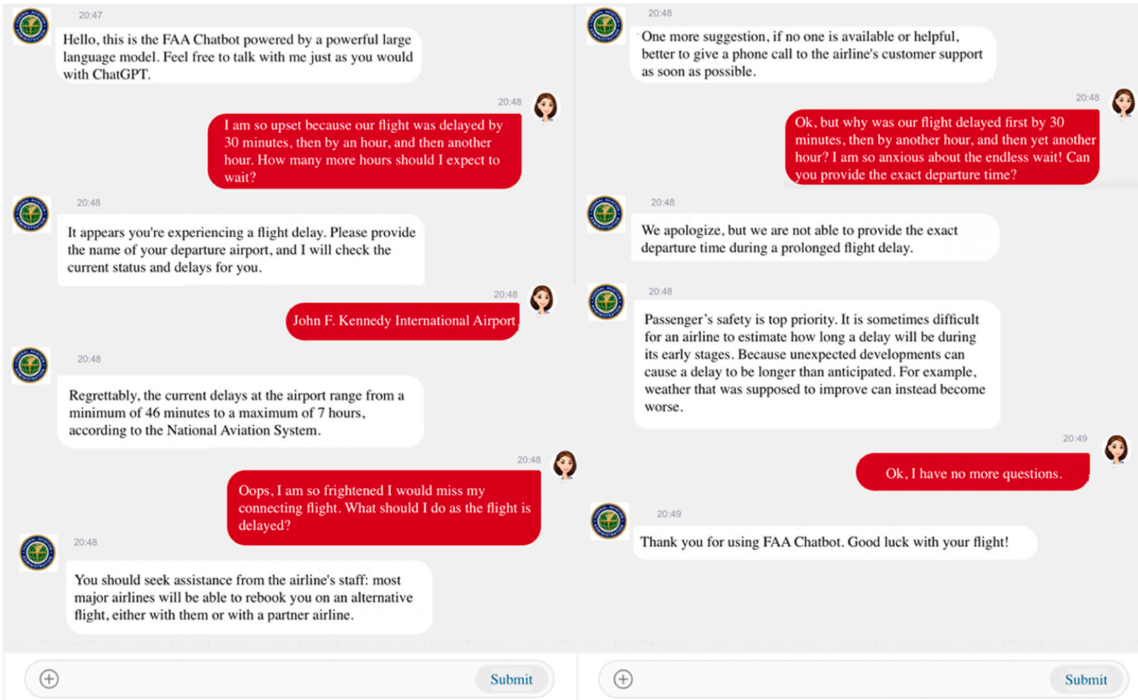
B. **Low competence * adjusting information**



C. **High competence * instructing information**



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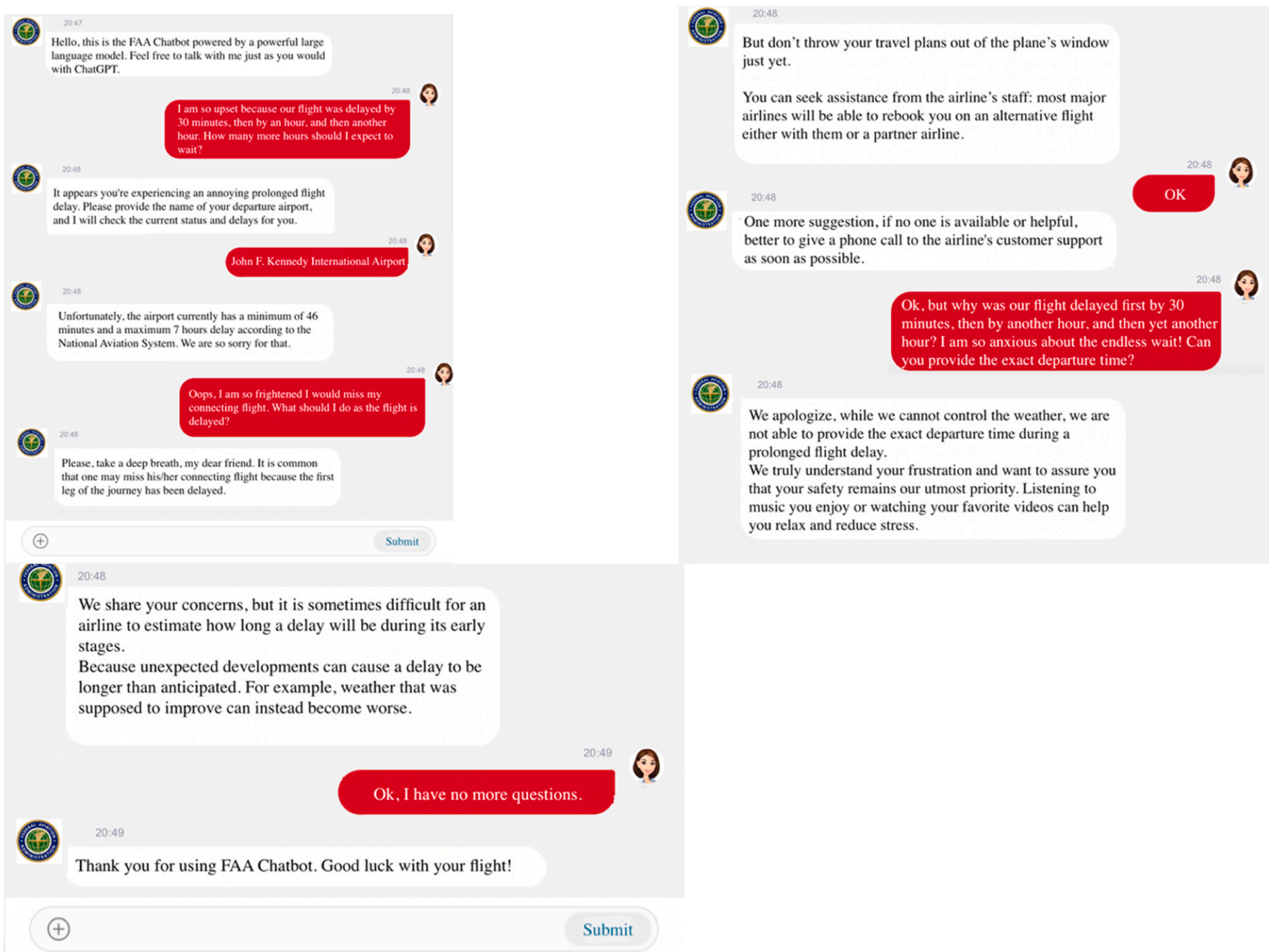
D. **High competence * Adjusting information**



.

Appendix F. Measurements of Study 2

a. Response message framing:
Instructing Information.

- FAA informed passengers about the delay.
- FAA guided individuals on how to manage their travel during the delay.
- FAA advised passengers on available amenities and services.
- FAA provided updates to keep passengers informed about the situation
- FAA provided updates to keep passengers informed about the situation.

Adjusting Information.

- FAA offered some emotional support to reduce anxiety caused by the delay.
- FAA delivered emotional support to the individuals affected.
- FAA extended emotional assistance to mitigate stress and frustration.
- FAA demonstrated its commitment to addressing passengers' concerns.
- FAA conveyed understanding and empathy for the uncertainty experienced by passengers.

b. Perceived competence of the agent.

- To what extent do you agree the online agent is skilful/ efficient/ competent/ intelligent/flexible?

c. Satisfaction with the organization.

- To what extent do you satisfy with the reaction from FAA on a seven-point Likert bipolar scale?
- Completely unacceptable to 7-completely acceptable
- Extremely bad to 7- extremely good
- Completely unsatisfied to 7-completely satisfied.

d. Attitude towards chatbot:

- Largely, I am fairly pleased with the chatbot services.
- Overall, the chatbot services fulfill my initial expectations.
- Generally, I am happy with the contents of the chatbot services.

Appendix G. : The Scenario for Study 3

Sarah is a resident of a region recently ravaged by a wildfire. Her community is in disarray – homes destroyed, power outages widespread, and a suffocating blanket of smoke has made life difficult. Evacuated from her home, Sarah is feeling lost and unsure where to turn for reliable information and support. She recalls hearing that the county's emergency response system has implemented a chatbot service. Accessing their website, Sarah locates the chatbot and initiates a conversation. Here's a breakdown of her interactions with the chatbot: Information Updates: The chatbot provides Sarah with regular updates on the wildfire situation. She receives details about the fire's current location, active evacuation zones, and the progress of containment efforts. Shelter Locations: Sarah needs to find a safe place to stay. The chatbot furnishes a list of operational evacuation shelters, complete with addresses, hours of operation, and details about the amenities offered at each site. Accessing Resources: With her medication running low due to pharmacy closures, Sarah requires assistance. The chatbot directs her to available emergency prescription services and mobile clinics operating within the affected area. Addressing Uncertainty: Beyond immediate needs, Sarah feels a deep sense of unease and needs reassurance. The chatbot provides basic guidance and connects her with resources for mental health support during the crisis.

Appendix H. : The procedure of the text analysis

1. LLM Selection: We selected three LLMs that can handle more than 12,000 tokens: GPT-4 Turbo, Claude 3 Sonnet, and Gemini 1.0 Ultra. These LLMs were chosen for their advanced language understanding capabilities and the ability to process longer text inputs.
2. Data Preparation: The qualitative data from the open-ended responses was collated and prepared for analysis. The text was merged into a single chunk and was ensured that the input did not exceed the token limit of the LLMs.
3. Prompt Design: A prompt was crafted to instruct the LLMs to identify the main topics present in the given text. The prompt included specific instructions for the LLMs to generate a structured output with three components for each identified topic: the title, definition, and number of occurrences. The following prompt was used:

//You are a qualitative researcher, and your task is to identify the topics in the text. When generating the topics, prioritize correctness and ensure that your response is accurate and grounded in the context of the text.

Your summary should include three components: the first one is the title of the topic; the second one is the definition of the topic; the third one is the number of occurrences of the topic.

insert topic 1: Title, definition, occurrence.
insert topic 2: Title, definition, occurrence.
insert topic 3: Title, definition, occurrence.
….
Here is the text://.

4. Topic Extraction: Each LLM (GPT-4 Turbo, Claude 3 Sonnet, and Gemini 1.0 Ultra) was prompted separately with the prepared text and the instructions to identify the main topics. The LLMs generated their respective outputs, listing the identified topics with their titles, definitions, and occurrence counts.
5. Consistency Evaluation: To evaluate the consistency of the topic extraction across the different LLMs, we calculated the coverage of topics by comparing the identified topics with those from prior research. Specifically, we divided the number of identical topics identified by the GPT 4 compared to the topics from Gemini and Claude. This provided a measure of the consistency and coverage of the topics extracted by the LLMs.
6. Topic Refinement: Discrepancies in topic granularity were addressed by comparing the outputs from the different LLMs. We fine-tuned the categorizations by merging related topics or splitting broad topics into more specific ones, ensuring coherence and alignment with the qualitative data.
7. Consistency Comparison: The comparison of the topic modeling results from the three LLMs revealed a notable consistency in the identified topics and their descriptions. While there were some variations in the granularity and labeling of topics, the overall themes and concerns raised by the participants were largely aligned across the three models.
8. Documentation: The detailed procedure, including the prompts used, the LLM outputs, the consistency evaluation process, the consistency comparison across LLMs, and the topic refinement steps, were documented for transparency and to facilitate comprehensive understanding of the analysis process.

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